

Jack of All Trades, Master of Some: Racial Competency and the Limits of Multiracial Representation*

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Abstract

Minority voters seek descriptive representation to achieve substantive outcomes, yet shared identity cannot guarantee the contextual knowledge group-specific policy demands. I update the agency framework to introduce domain-specific competency into the incumbent's behavioral calculus, providing a general approach to group-based substantive representation. I apply this dimension of accountability to racial dynamics by introducing racial competence. Multiracial politicians serve as a unique case, as their framing as descriptive representatives of multiple groups leaves their racial competence in question. Results qualify the notion that a Multiracial identity provides a strategic advantage: voters' perceptions of racial competence drive incumbents to posture, risking exposure when future periods demand contextual knowledge. Extensions show how electoral and social institutions alter the incumbent's behavior. Racial gerrymandering amplifies posturing and lowers voter welfare, while caucus membership provides insurance against future demands for competence. Domain-specific competency provides a framework for examining formal extensions and empirical implications of Multiracial behavior.

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“I don’t think you can consider the issue of mixed race outside of the issue of race. I think it’s important to try to avoid that sense of exclusivity . . . Don’t ever think of yourself as so unique that you divorce yourself from your communities.”

— Barack Obama (2005)

Introduction

What it means to represent minority interests in America remains a contested issue. Representation is not only about shared identity but also the ability to act effectively for minority constituents through shared experiences and contextual knowledge. By electing coracial politicians, voters reduce the information gap and place greater trust in their representatives.¹ Yet, whether descriptive identity is sufficient to ensure credible action remains unresolved, making domain-specific competency the central tension to understanding how minority representation works.

A plethora of agency models within the political economy tradition highlight the role competency plays in structuring an incumbent’s behavior (Bils 2023; Canes-Wrone, Herron, and Shotts 2001; Fox 2007; Maskin and Tirole 2004). However, this class of models cannot speak to cases in which voters demand domain-specific competency or substantive knowledge of issues about a particularized group. Here, I update the agency framework to introduce domain-specific competency into the incumbent’s behavioral calculus, providing a general approach to questions of group-based substantive representation. Importantly, I apply this dimension of accountability to racial dynamics by introducing *racial competence*. A politician with shared experiences is racially competent and can credibly navigate issue domains that are salient to both racial minorities and majority groups. In contrast, when the policy domain requires contextual knowledge and a politician lacks shared experience, that politician lacks such competence.

My model provides a pathway to understanding how voters’ perceptions of racial competence drive the incumbent’s behavior. I introduce contextual knowledge as a probability distribution over the policy domain, which can be either universal or particularized. The incumbent officeholder faces uncertainty about the need for racial competence in the next period. Importantly, racial competence is conditional on the policy domain. Voters are likewise uncertain about the incumbent’s ability to act with racial competence. Each period, the incumbent selects a policy based on their type and domain, and voters decide whether to reelect or replace them. This two-sided uncertainty is sharpest when a politician’s identity does not clearly signal shared experience, and no group of politicians embodies that ambiguity more than Multiracials.

1. Race serves as a heuristic for how well a politician can represent minority voters by signaling perceived co-racial alignment, particularly in low-information elections (Dawson 1994; McConaughy et al. 2010).

Multiracial politicians offer a unique opportunity to examine this dynamic, as their choice to identify as mixed-race in political institutions raises questions among minority voters about their commitment to group-specific policy goals.² Americans are increasingly identifying as mixed-race, and Multiracial politicians now comprise a growing minority of incumbents (Hardy-Fanta et al. 2013; U.S. Census Bureau 2022).³ Multiracial politicians are framed as the “vaccine” to inter-racial conflict in the United States, in part because of their perceived ability to navigate shared experiences with multiple racial constituencies (Lemi 2021; Velasquez-Manoff 2017).⁴ Politicians of mixed-race ancestry have likewise promoted the idea that their racial identity is strategic, enabling them to connect with multiple groups (Brown 2014; Lemi 2018). As a result, the stakes of correctly assessing their racial competence are even higher. Yet these same strategies create uncertainty about their competence, as constituents doubt their ability to act in situations that demand contextual knowledge.

The interaction between voter perceptions and Multiracial politicians’ attempts to demonstrate competence creates a feedback dynamic: on one hand, voters seek signals of racial competence to ensure their group-specific policy needs are addressed (Wamble 2025). Multiracial politicians, in turn, participate in identity-specific caucuses, engage in symbolic forms of representation, and support legislation benefiting minority interests to demonstrate competence (Lemi 2018). Yet because voters cannot easily place Multiracial politicians as descriptive representatives, they often prefer in-group monoracial candidates instead (Lemi 2021).⁵ This dynamic makes perceptions and actions interdependent. Multiracial politicians act to shape voter perceptions, while voter perceptions simultaneously condition which actions are seen as credible displays of racial competence.

This dynamic is at the core of my model. Voters want a politician who can deliver when the issue of the day demands it, but are uncertain whether the incumbent can do so; incumbents want to be perceived as competent but face uncertainty about their ability. As a result, incumbents have incentives to distort policy to appear racially competent. More broadly, unpacking this interplay is essential for understanding how politicians’ behavior varies across constituencies and settings with diverse policy needs.

However, the difficulty in this interplay is that it masks a principal-agent problem. In many constituencies, coracial alignment appears clear—because a voter shares observable characteristics with their representative, their interests are assumed to be aligned. Yet, as

2. I focus on Black-White Multiracial politicians, whose identities are especially salient given their shared racial history in the U.S. In particular, the legacy of hypodescent and the one-drop rule provides clear institutional and social incentives for Black-White Multiracials to align with monoracial Black interests (Davis 1991).

3. However, see Starr and Pao (2024), who argues the 2020 spike reflects changes in measurement rather than a sudden demographic shift.

4. Pew (2015) found that one-in-five Multiracial Americans view themselves as a bridge between racial groups. Similarly, Hochschild and Weaver (2010) notes growing support for mixed-race heritage as a means of tearing down racial barriers.

5. Lemi (2021) also shows that respondents are more supportive of mixed-race politicians when they have a shared identity than those who do not, highlighting the fact that any descriptive representation is better than none. This finding also provides evidence for the role shared experiences play in voters’ evaluations of Multiracial politicians.

Wamble (2025) and Dovi (2002) emphasize, shared identity does not necessarily translate into racial competence. Competence is unobservable to constituents and can only be inferred through a politician’s actions. This creates space for distortion, as politicians may alter their behavior to shape perceptions of their racial competence. Multiracial politicians, in particular, provide variation in shared identity that untethers the link between descriptive and substantive representation.

The game is structured such that the incumbent is a Multiracial politician. As such, competency here is less about a politician’s general ability to handle a crisis (Bils 2023) and more about shared experiences and how these translate into domain-specific knowledge. The asymmetric information environment aligns with the idea that voters are also unaware of the incumbent’s in-group status. Because Multiracial politicians intentionally select a mixed-race identity (Brown 2014; Lemi 2018), voters have to acquire information about their level of racial competence through their policy decisions.⁶ Three main takeaways from the model are evident.

First, I qualify the notion that maintaining a Multiracial identity provides a strategic advantage. Baseline analyses indicate that when electoral incentives are strong, Multiracial politicians exaggerate policy to secure reelection. Yet, this risks the politician being “outed” as incompetent, particularly if the future demands contextual knowledge and voters’ prior beliefs are that the incumbent is racially competent. Thus, the feedback dynamic is clear: voters’ perceptions of the Multiracial politicians’ racial competence drive the incumbent’s incentive to be regarded as having domain-specific knowledge.

Racial gerrymandering serves as a core application of my model, with both behavioral and welfare implications. As the probability that a district is cracked (introducing racial heterogeneity among voters) increases, Multiracial politicians are more willing to posture in equilibrium. Significantly, this finding resolves a core tension in the empirical gerrymandering literature by highlighting a behavioral effect of cracking on Multiracial incumbents (Cameron, Epstein, and O’halloran 1996). Voter welfare is also affected by gerrymandering. By splintering voters who require racial competence, cracking diminishes the accountability mechanism necessary for preventing policy distortion. I show that while there may be little perverse effects along an ideological dimension (Shotts 2001, 2002), racial gerrymandering impacts minority voters’ accountability.

Finally, caucus membership and the constraints imposed serve as an important conduit of information for both the incumbent and the voter. Multiracial incumbents are keen on participating in identity-based organizations within the legislature (Lemi 2018). Part of the reason is that constraining yourself via caucus membership serves as insurance against future periods where racial competence is required. Moreover, caucus membership provides the incumbent with opportunities to convey contextual knowledge to their constituency through leadership positions and symbolic representation (Canon 1999). Caucus constraints

6. Individuals who self-identify as Multiracial are also more likely to have been socialized in predominantly non-minority spaces (Rockquemore 1999; Rockquemore and Brunsma 2002b; Root 1990), directly impacting their ability to acquire contextual knowledge through lived experiences.

also serve as an important social monitor, forcing the Multiracial incumbent back in line with coracial policy goals (Rendleman 2023). This mechanism relates more broadly to racialized social constraints and Multiracial behavior at the mass level (White and Laird 2020).

This paper contributes to the study of Multiracialism by formalizing how the choice to identify as Multiracial generates uncertainty about racial competence. Existing theories of Multiracial politicians examine voter evaluations of mixed-race candidates (Lemi 2021; Leslie et al. 2022).⁷ The core aspect of this literature is how voters perceive Multiracial politicians. Multiracial politicians complicate the racial status quo, bridging inter-racial coalitions while alienating coracial constituencies (Lemi 2021; Leslie et al. 2022). Multiracial politicians are acutely aware of how voters perceive them, with their credibility depending on both their bridge-building appeal and on how convincingly they demonstrate racial competence to their community (Lemi 2021). Multiracial politicians often perceive their racial identity as an asset, believing it allows them to appeal across constituencies (Brown 2014). There are clear incentives for Multiracial incumbents to choose how to identify to their constituency (Lemi 2018, 2026). Yet, neither strand formalizes how voters' perceptions of racial competence structure the actions of Multiracial politicians. My model directly addresses this gap by showing how electoral incentives condition when Multiracial incumbents choose to exaggerate policy or reveal competence, clarifying the strategic tradeoffs they face between appealing broadly as a coalition builder and signaling credibility to coracial voters.

By allowing for variation in domain-specific competency, my model addresses whether descriptive representation translates into substantive benefits for minority voters by exploiting the strategic choice of identifying as Multiracial in electoral politics. Extant work discusses how descriptive representation leads to substantive outcomes for minority constituents because voters look for signals of racial competence (Dovi 2002; Mansbridge 1999; Wamble 2025; Williams 1993).⁸ Young (1997), however, challenges this by noting that racial competence is not guaranteed by a singular group identity in the descriptive sense but is instead conditioned by one's social positioning. No single identity or actor can represent all interests; instead, social positioning opens the possibility for multiple actors to engage in the same perspective (Young 1997). On one hand, Multiracial representatives embody Young's proposition by condensing multiple social positions into a single actor, thereby potentially offering a plurality of perspectives at once. On the other hand, my model suggests that this multiplicity complicates the very logic of descriptive representation, as Multiracial politicians struggle to credibly demonstrate racial competence to their constituents. Even when Multiracial politicians are reelected, they may have distorted policy through purely symbolic means. And while I do not debate the value of symbolic representation (Tate 2004), if the framing of Multiracial politicians is that they can bridge racial divides by meeting the needs

7. Mixed-race Americans face many mechanisms that drive identification (Rockquemore and Brunsma 2002a). One mechanism, reflected appraisals, holds that Multiracial identity is driven by how others perceive them (Khanna 2004, 2010; Sims 2016). These perceptions are fluid and influenced by the actions taken by mixed-race Americans (Sims 2016).

8. while some scholars have argued that descriptive representation is neither necessary nor sufficient for substantive policy outcomes (Pitkin 1967; Swain 1995), others have demonstrated the clear implications of having coracial voices in Washington (Butler and Broockman 2011; Grose 2011; Haynie 2001; Hero and Tolbert 1995; Minta and Brown 2014; Preuhs 2006; Whitby 2000).

of multiple constituencies, my analyses depict a more complicated reality.

More broadly, my model extends existing work in the political economy literature (Bils 2023; Canes-Wrone, Herron, and Shotts 2001; Fox and Stephenson 2011; Maskin and Tirole 2004) by treating descriptive representation as an agency problem. Unlike prior accounts, which assume competence is exogenous (Bils 2023; Canes-Wrone and Shotts 2007), I demonstrate how perceptions of competence are shaped by a politician’s ability to demonstrate domain-specific competency. This creates a feedback dynamic between voters’ assessments of credibility and politicians’ strategic re-election-seeking behavior. Quality is now *endogenous to the policy domain*. When applied to racial dynamics, this demonstrates how domain-specific competency influences whether minority voters perceive politicians as racially competent. By situating racial competence at the center of political economy models, this paper highlights how minority communities’ perceptions actively condition representatives’ behavior.

In this respect, the underlying mechanism of my model is most closely related to Bils (2023), which also introduces competency as a guiding force in incumbents’ decision-making. There are two mechanisms by which incumbents distort policy: anti-herding (or overreacting) (Levy 2004; Prendergast and Stole 1996), and posturing (or exaggerating) (Fox and Stephenson 2011). Incumbents exaggerate policy to appear competent when they lack the necessary information to make informed decisions. Yet what remains unclear from Bils (2023) is how shared experiences condition competency. For minority voters, navigating a crisis requires racial competence, yet this community-specific dimension remains unaddressed (Dovi 2002; Mansbridge 1999; Williams 1993). My framework provides an avenue for incorporating this idea, situating racial competence at the center of accountability models for minority constituencies.

The rest of the article is structured as follows. First, I introduce the preliminaries of the model, discussing the players, actions, timing, information, and equilibrium concept. Next, I present the results for the baseline version of the model, highlighting the role of information in structuring behavior. In extensions, I discuss a core application of my model to racial gerrymandering and the role legislative caucuses play in constraining incumbent behavior. Finally, I conclude with a synopsis of the findings and implications for empirical work.

Model Description

Players. There are three players in the model: a Multiracial incumbent (I) who is seeking reelection, a challenger (C), and a Black voter (V).⁹

9. The race of the challenger is intentionally blank as welfare analyses highlight how one can interpret the voter’s prior beliefs about the challenger’s racial competence as a proxy for their racial background.

Timing. The model has two periods, $t \in \{1, 2\}$. In each period, an incumbent chooses a policy $x_t \in X = \mathbb{R}$. At the end of the first period, a voter observes x_t and chooses to reelect the incumbent or elect the challenger. I exclude legislative caucus constraints from the baseline model to isolate the core role that racial competence plays in influencing policy distortion.

Preferences. In the baseline model, players have the same policy preferences, represented by an ideal point ω_t . The voter’s utility is quadratic loss and is represented by $-(x_t - \omega_t)^2$. The incumbent’s utility is

$$-(x_t - \omega_t)^2 + \mathbb{1}_t \beta,$$

where $\beta \in (0, \infty)$ is the office benefit a politician receives for each period they hold office, and $\mathbb{1}_t$ indicates whether the politician is in office. Politicians aim to maximize their utility over the two periods.¹⁰

Information. In each period, nature randomly determines the state of the world, ω_t . The state of the world represents the ideal policy position and is unknown to voters and uninformed incumbents. Let ω_t be drawn from a known distribution F , which has a probability density f . Furthermore, let F have full support over $\mathbb{R}$ and be distributed with a mean of 0 and $\sigma^2 > 0$. Finally, I assume that ω_1 and ω_2 are uncorrelated.

A politician’s competence is conditional on the policy domain. I assume that an officeholder is aware of the current period’s domain when making policy choices; however, the future domain is uncertain. I introduce a new parameter, ρ , which captures the probability distribution over policy domains. Specifically, the policy domain d_t is drawn by nature with probability $\rho = \Pr(d_t = U)$ for the universal domain (U), and $1 - \rho = \Pr(d_t = P)$ for the particular domain (P). This addition highlights that electoral contexts are not homogeneous. The environment in which a politician operates may emphasize either broad, universal concerns or group-specific, particularized concerns.

Whether a politician is viewed as racially competent depends on the policy domain. Let a politicians’ type be defined as $\theta_j \in \{H, L\}$ for $j \in \{I, C\}$. Informed types (H) are fully aware of ω_t and are competent across both domains. Uninformed types (L) are competent only in the universal domain; when placed in the particular domain, they are unable to internalize ω_t . Uncertainty about type θ_j is represented by a common prior that a politician is racially competent with probability $\pi_j \in (0, 1)$.

The core contribution is that racial competence is no longer viewed as an inherent trait (Bils 2023), but rather as something that depends on the policy environment. This captures

10. In Section D of the Supplementary Materials, ideology is included as an additive parameter in the incumbent’s utility to understand how racial competence interacts with polarization.

how minority representation is contingent not just on whether an official is “competent” in the abstract, but on whether their competence aligns with the domain-specific demands of their constituency. Within the baseline and in extensions, racial competence plays a significant role through information, impacting not only the incumbent’s behavior but also voter welfare.

Equilibrium. I study perfect Bayesian equilibria (PBE). Specifically, let $\mu : X \rightarrow [0, 1]$ represent a mixed strategy for the voter, where $\mu(x)$ is the probability of reelecting the incumbent following their policy choice x . Strategies are sequentially rational given beliefs, and beliefs are derived from Bayes’ Rule whenever possible. For the politician, a mixed strategy is defined by $\sigma_t : \mathbb{R} \cup \{\phi\} \rightarrow \Delta(X)$, where σ_t maps their private information (informed $\mathbb{R}$ or uninformed ϕ) to a probability distribution over X . To restrict off-path beliefs, I focus on PBE satisfying D1 (Cho and Kreps 1987). Intuitively, upon observing an off-path policy, voters’ beliefs would assign a higher probability to the incumbent type that has more to gain from deviating.

Results

The results section is organized as follows. In the baseline model, I analyze how uncertainty about an officeholder’s racial competence shapes equilibrium policymaking and electoral outcomes. For voters, this is apparent through their decision rule. For incumbents, racial competence affects both the extremity and willingness to distort policy, improving their probability of reelection. Voter welfare analysis shows that perceptions are the primary driver of an officeholder’s willingness to distort policy. In extensions, racial gerrymandering modifies the incumbent’s willingness to posture while diminishing voter welfare. At the same time, uncertainty about racial competence shows how caucus membership acts as a safeguard against the need for contextual knowledge in future periods.

Baseline Findings

In the second period, there is no electoral incentive, so the officeholder selects their optimal policy based on their information about ω_2 . The voter’s continuation value from electing the challenger depends on three main components: the probability that the challenger is uninformed ($1 - \pi_C$), the probability that the current period requires contextual knowledge ($1 - \rho$), and the policy loss from the low-type selecting $x_1 = 0$ ($-\sigma^2$). Formally, the voter’s continuation value from reelecting the incumbent is $V(I \mid x_1) = -[1 - \tilde{\pi}_I](1 - \rho)\sigma^2$, while the value of electing the challenger is $V(C) = -[1 - \pi_C](1 - \rho)\sigma^2$.

Reducing the need for racial competence raises the voter’s continuation value, as higher values of ρ increase the expected payoff from electing the challenger. Lemma 1 below estab-

lishes this result.

Lemma 1 (Impact of ρ on Continuation Values). *The voter’s continuation value from electing the challenger is strictly increasing in ρ , and converges to 0 as ρ approaches 1.*

Lemma 1 shows that the policy domain directly conditions the voter’s continuation value by reducing the risk of policy loss from an uninformed challenger. As ρ approaches 1, the need for racial competence decreases, and the voter’s decision rule is determined entirely by the incumbent’s first-period behavior. Conversely, as ρ approaches 0, the need for racial competence increases, and the voter compares the incumbent’s posterior probability of being competent to the challenger’s prior. An analogous condition applies to the incumbent’s continuation value.¹¹ Lemma 1 also impacts the voter’s decision rule by collapsing the continuation values from both the incumbent and challenger to 0 as ρ approaches 1. Therefore, the posterior belief about the incumbent is only significant to the voter’s calculus when $\rho < 1$.

Next, to define the voter’s decision rule, sequential rationality requires that the voter select the candidate that provides the greatest continuation value, as there is no commitment mechanism. Let $\tilde{\pi}_I$ be the voter’s posterior belief that the incumbent is competent. Then the voter’s decision rule is determined by the posterior belief that the incumbent is racially competent and the prior belief about the challenger.

Lemma 2 (Voter Decision Rule). *After observing the first-period policy x_1 in equilibrium, the voter (1) reelects the incumbent if $\tilde{\pi}(x_1) > \pi_C$, (2) reelects the challenger if $\tilde{\pi}(x_1) < \pi_C$, and (3) is indifferent if $\tilde{\pi}(x_1) = \pi_C$.*

In equilibrium, Lemma 2 shows that a comparison of continuation values pins down the condition on beliefs. If $V(I | x_1) > V(C)$, the voter will reelect the incumbent. Similarly, if $V(I | x_1) < V(C)$, they will elect the challenger. When $V(I | x_1) = V(C)$, the voter mixes and re-elects the incumbent with some probability $\mu(x_1) \in [0, 1]$. Importantly, racial competence enters into the voter’s decision rule through their continuation values. As such, racial competence directly affects voters’ posterior and prior beliefs. This will be useful in an extension examining voter welfare.

Complete Information Benchmark

I now characterize first-period behavior under complete information. This baseline illustrates electoral incentives when the voter’s uncertainty about type is removed. Voters are fully aware of the incumbent’s type, so the election is a foregone conclusion. Racial competence structures the incumbent’s optimal action as follows: when $d_1 = U$, both types select $x_1 = \omega_1$; conversely, when $d_1 = P$, the competent type matches the state while the incompetent incumbent chooses $x_1 = 0$. This is analogous to Bils (2023) with the critical departure being

11. All proofs are provided in the Supplementary Materials.

that the second-period policy setting is unknown, creating additional uncertainty for the low-type. Remark 1 summarizes the equilibrium behavior.

Remark 1. *Under the complete information benchmark, there is a subgame perfect Nash equilibrium in which the high-type incumbent always selects $x_1 = \omega_1$, and the low-type incumbent selects*

$$x_1 = \begin{cases} \omega_1, & \text{if } d_1 = U, \\ 0, & \text{if } d_1 = P. \end{cases}$$

The voter always reelects the high-type and rejects the low type.

Remark 1 helps ground the role of racial competence in both voters' perceptions and incumbents' actions. Here, voters know the competent incumbent; as a result, the uninformed type has no incentive to distort policy to be perceived as competent. However, as the next section shows, when voters cannot observe the incumbent's type, politicians strategically exaggerate policy when office rents are high. The degree of distortion is structured by racial competence.

Uncertainty about racial competence

I now introduce voter uncertainty about the incumbent's racial competence. Under incomplete information, incompetent types may be willing to adopt a *posture* and, in equilibrium, select policy positions that force the informed incumbent to *overreact* to signal their competence to voters. Specifically, following Bils (2023),

Definition 1 (Overreacting and Posturing). *An informed incumbent overreacts when $\omega > 0$ and they choose $x > \omega$, or when $\omega < 0$ and they choose $x < \omega$ (Levy 2004; Prendergast and Stole 1996). An uninformed incumbent postures when they choose any $x \neq \mathbb{E}[\omega] = 0$ (Fox and Stephenson 2011).*

I assume that the incumbent is aware of the policy domain in the first period.¹² Because racial competence is not required when $d_1 = U$, both incumbent types match the state. As a result, I examine first-period behavior when $d_1 = P$ unless noted otherwise. To introduce the dynamics of this section, note that the high-type never deviates from equilibrium behavior, as they know the state in each period. Therefore, the incentive for the uninformed incumbent to deviate guides first and second-best behavior. Let the incentive compatibility constraint for the uninformed type be

12. It is safe to assume that politicians are aware of their constituents' present policy needs by performing constituency service, yet still unaware of what the future holds (Fenno 1978). For example, while a minority community may not require particularized expertise in the current period, a racialized event may necessitate it tomorrow. Future research could expand the setting to a model in which particularized interest groups also demand racial competence from elected officials.

$$\underbrace{\beta - \sigma^2}_{\text{Equilibrium Payoff}} - \underbrace{[1 - \pi_C]\sigma^2}_{V(C)} = \underbrace{2\beta - (2 - \rho)\sigma^2}_{\text{Total Expected Payoff}}. \quad (1)$$

where the equilibrium payoff is the uninformed type's expected utility from choosing $x_1 = 0$ and getting kicked out of office. The total expected payoff is if the uninformed type postured and selected a different policy, resulting in their reelection. This is analogous to Bills (2023) except for the uncertainty ρ introduces in the second period. Solving for this provides the office rent cut-points that delineate two cases: low-value office, when $\beta \leq \pi_C\sigma^2 - \rho\sigma^2$ and high-value office, where $\beta > \pi_C\sigma^2 - \rho\sigma^2$.

Low-value Office

When office rents are low, uninformed incumbents face no incentive to posture, as the risk of policy loss from deviating is too great. Therefore, they select their optimal first-best outcome, $x_1 = 0$, and are kicked out of office. Remark 2 highlights this finding, which is standard in accountability models with competence considerations (Bills 2023; Canes-Wrone, Herron, and Shotts 2001).

Remark 2. *If office rents are low so that $\beta \leq \pi_C\sigma^2 - \rho\sigma^2$, then there exists a perfect Bayesian equilibrium in which the high-type incumbent selects $x_1 = \omega_1$ and the low-type incumbent selects $x_1 = 0$.*

Under this condition, types fully separate in equilibrium; voters update their beliefs with probability 1 that the incumbent is racially competent, ensuring they reelect the informed type and reject the uninformed type. Substantively, when office benefits are low, an uninformed incumbent prefers to accept defeat rather than posture to distort beliefs about their commitment to delivering on particularistic policy.

High-value Office

Suppose now that the office benefits are high ($\beta > \pi_C\sigma^2 - \rho\sigma^2$). In this case, the uninformed type has an incentive to *posture to appear racially competent*. To characterize this tradeoff, assume there exists some policy choice $-x^2$ that leaves the uninformed type indifferent between following the equilibrium strategy and deviating. Following Bills (2023), the incentive compatibility condition is

$$\underbrace{\beta - \sigma^2}_{\text{Equilibrium Payoff}} - \underbrace{[1 - \pi_C]\sigma^2}_{V(C)} = \underbrace{-x^2}_{\text{Policy Indifference}} + \underbrace{2\beta - (2 - \rho)\sigma^2}_{\text{Total Expected Payoff}}. \quad (2)$$

Solving yields the policy bounds $\underline{x}^P(\rho)$ and $\bar{x}^P(\rho)$ at which the uninformed incumbent is indifferent: $\bar{x}^P(\rho) = \omega^P(\rho) = \sqrt{\beta - \pi_C \sigma^2 + \rho \sigma^2}$. At this point, the model follows Bils (2023) in using incentive compatibility to locate the thresholds that discipline behavior. The critical departure, however, lies in the uncertainty surrounding the second period. Uncertainty about the need for racial competence affects the incumbent's willingness and the extent of policy extremity. In earlier models, the environment is fixed, so distortion depends only on office rents and type uncertainty. Here, the possibility that voters may require domain-specific knowledge in the next period directly enters the cut-point through the $\rho \sigma^2$ term. This modification shifts both *how far* the uninformed type will posture and *how much* the informed type must exaggerate to signal competence.

The probability that ω lies between 0 and $\omega^P(\rho)$ determines the incumbent's willingness to distort policy (equivalently, the lower bound lies between $-\omega^P(\rho)$ and 0). At the boundary $x_1 = \bar{x}^P(\rho)$, the voter is indifferent and by Lemma 2, re-elects the incumbent with some probability $\mu(x_1) \in [0, 1]$. Accordingly, the cut-points defining the probability that ω_1 lies within the threshold range are

$$\bar{\Sigma} = \frac{\pi_I(1 - \pi_C)}{\pi_C(1 - \pi_I)}(F(\omega^P(\rho)) - F(0)), \quad \underline{\Sigma} = \frac{\pi_I(1 - \pi_C)}{\pi_C(1 - \pi_I)}(F(0) - F(-\omega^P(\rho))).$$

Here, racial competence enters through the state threshold, amplifying the mixing probability for the uninformed incumbent as well as their policy extremity. With that in mind, I characterize the set of Perfect Bayesian Equilibria of the model.

Proposition 1 (Perfect Bayesian Equilibria with High Office Rents). *The set of Perfect Bayesian Equilibria of the model is characterized as follows:*

1. Voter strategy.

- If $x_1 \in (\underline{x}^P(\rho), \bar{x}^P(\rho))$, the voter rejects the incumbent.
- If $x_1 \leq \underline{x}^P(\rho)$ or $x_1 \geq \bar{x}^P(\rho)$, the voter re-elects the incumbent.

2. Informed incumbent.

- If $\omega_1 \in [0, \omega^P(\rho))$, the informed type overreacts by setting $x_1 = \bar{x}^P(\rho)$.
- If $\omega_1 \in (-\omega^P(\rho), 0]$, the informed type overreacts by setting $x_1 = \underline{x}^P$.
- If $|\omega_1| > \omega^P(\rho)$, the informed type plays the first-best by choosing $x_1 = \omega_1$.

3. Uninformed incumbent.

- With probability $\bar{\sigma} \in [0, \min\{1, \bar{\Sigma}\}]$, the uninformed type postures right by choosing $x_1 = \bar{x}^P(\rho)$.
- With probability $\underline{\sigma} \in [0, \min\{1 - \bar{\sigma}, \underline{\Sigma}\}]$, the uninformed type postures left by choosing $x_1 = \underline{x}^P(\rho)$.

- With residual probability $1 - \underline{\sigma} - \bar{\sigma}$, the uninformed type plays the first-best by setting $x_1 = 0$.

Off the path of play, assume the voter updates their belief with probability 1 that the incumbent is uninformed.

Under Proposition 1, the incumbent distorts policy to signal racial competence. If $x_1 \in (\underline{x}^P(\rho), \bar{x}^P(\rho))$, the voter infers that the incumbent is uninformed and turns to the challenger. By contrast, when policy is set at or beyond the thresholds $|x_1| > \bar{x}^P(\rho)$, the voter updates with probability 1 that only an informed type would choose such a position, and reelects the incumbent. In this region, the uninformed type can mix with probability $\Sigma \in [\underline{\Sigma}, \bar{\Sigma}]$. As long as the uninformed type does not mimic too often, the voter remains willing to retain the incumbent. An analogous result is evident for $x_1 = \underline{x}^P(\rho)$. Figure 1 illustrates this characterization.

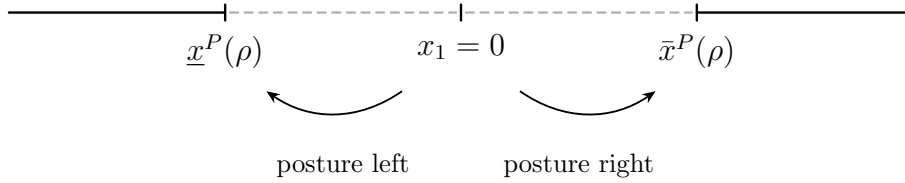


Figure 1: Posturing by the uninformed incumbent. The moderate policy $x_1 = 0$ costs the incumbent re-election, and any position strictly between the cut-points $\underline{x}^P(\rho)$ and $\bar{x}^P(\rho)$ (dashed) is seen through by the voter. To survive, the uninformed incumbent abandons $x_1 = 0$ and postures to $\underline{x}^P(\rho)$ or $\bar{x}^P(\rho)$.

This structure parallels Bilal (2023), in which informed incumbents exaggerate to distinguish themselves, and uninformed incumbents mix to exploit this exaggeration. However, because an incumbent's behavior is structured not only by their type but also by the need for contextual knowledge, my model expands the range of actions an incumbent can take to distort policy. In that respect, Bilal (2023) becomes a special case where $\rho = 0$. Here, uncertainty alters the incentives on both sides, expanding the range in which uninformed types can posture and raising the level at which informed types must exaggerate to be believed. For the informed type, behavior depends on the state. When the state is extreme relative to the threshold, the informed type chooses their first-best policy. But when the state is more moderate and close to zero, they cannot simply match ω , since voters reject any policy choice inside the interval $(\underline{x}^P(\rho), \bar{x}^P(\rho))$. To demonstrate competency, the informed type exaggerates, selecting a boundary policy such as $x_1 = \bar{x}^P(\rho)$. The uninformed type mixes between posturing at the cutoffs, $x_1 = \bar{x}^P(\rho)$ or $x_1 = \underline{x}^P(\rho)$, and sometimes setting $x_1 = 0$. Because they cannot know the state ex ante, their survival depends on being seen as competent often enough to survive into the next period.

Proposition 1 connects directly to the feedback dynamic introduced at the start of the paper. Multiracial incumbents, whose descriptive ties are questioned, are especially pressured to distort policy to be perceived as racially competent. Because their ambiguity causes even competent incumbents' coracial status to be questioned, they feel the need to overreact

and exaggerate policy to demonstrate racial competence. For uninformed incumbents, this same ambiguity creates opportunities to slip through the cracks and be perceived as racially competent despite lacking the ability to deliver particularistic policy. In this sense, voters' perceptions amplify the incentives on both sides of the signaling game. It raises the cost of credible separation for the competent type, while expanding the scope for opportunism by the uninformed. The discussion section returns to this point, situating the model's logic within broader debates on race and representation.

To rule out other potential equilibria, I refine Proposition 1 using the set of non-off-the-path beliefs (Cho and Kreps 1987). The next remark demonstrates this refinement.

Remark 3. *The equilibria identified in Proposition 1 survive D1.*

Any policy choice $x \in (\underline{x}^P(\rho), \bar{x}^P(\rho)) \setminus \{0\}$ is off the path of play. Both types benefit from selecting $x \in (\underline{x}^P(\rho), \bar{x}^P(\rho))$. However, the uninformed incumbent's utility from deviating is larger than that of the informed type. Therefore, D1 requires the voter to update that the incumbent is the uninformed type with probability 1. This finding is equivalent to Bils (2023).

Contextual knowledge, ρ , enters the uninformed type's calculus through their upper and lower policy bounds. As ρ increases, the incentive to posture at the cut-point grows, since voters are less focused on racial competence and more willing to accept exaggerated policies as credible signals. Proposition 2 characterizes this comparative static by showing how ρ shifts the extremity of equilibrium policy choices.

Proposition 2 (Impact of ρ on Policy Extremity). *An increase in the probability that the second-period policy domain is universal leads to an increase in the extremity of the policy choice; that is,*

$$\frac{\partial \bar{x}^P(\rho)}{\partial \rho} > 0 \quad \text{and} \quad \frac{\partial \underline{x}^P(\rho)}{\partial \rho} < 0.$$

Furthermore, as $\rho \rightarrow 1$,

$$\lim_{\rho \rightarrow 1} \bar{x}^P(\rho) = \sqrt{\beta - (\pi_C - 1)\sigma^2} \quad \text{and} \quad \lim_{\rho \rightarrow 1} \underline{x}^P(\rho) = -\sqrt{\beta - (\pi_C - 1)\sigma^2}.$$

Proposition 2 shows that uncertainty about the need for domain-specific competency directly conditions the extremity of incumbents' actions. As the probability of a universal second period increases, the cut-points that discipline uninformed types shift outward, making more extreme policies credible. Figure 2 illustrates this comparative static. The policy line shows the positions available to an uninformed incumbent; $\underline{x}^P(\rho)$ and $\bar{x}^P(\rho)$ bound the semi-pooling region of the PBE, and the ρ effect pushes those bounds outward to $\underline{x}^U(\rho)$ and $\bar{x}^U(\rho)$. Substantively, when voters are more confident that the future domain is universal, it becomes easier for an uninformed incumbent to posture as racially competent. In effect, Multiracial incumbents are more willing to select extreme policies, and voters are more willing to reelect them in equilibrium. Proposition 3 formalizes this result.

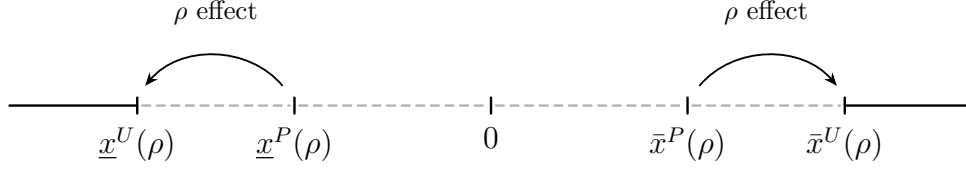


Figure 2: The ρ effect. As ρ rises, the voter expects racial competency to matter less tomorrow, so more extreme stands become credible. The cut-points shift outward from $\underline{x}^P(\rho)$, $\bar{x}^P(\rho)$ to $\underline{x}^U(\rho)$, $\bar{x}^U(\rho)$. The uninformed incumbent postures further from 0 and wins re-election more often.

Proposition 3 (Impact of ρ on the Winning Probability for the Uninformed Incumbent). *As ρ increases, the winning probability for the uninformed incumbent increases.*

For Multiracial incumbents in particular, Propositions 2 and 3 demonstrate the constraints that contextual knowledge imposes on the actions chosen in equilibrium. When voters require racial competence in the next period, Multiracials condition their posturing on these perceptions, knowing that they are unable to meet the demands of their constituency. However, when racial competence is not required, Multiracial incumbents are more willing to set policy at a higher (lower) bound to be perceived as the competent type.

Voter Welfare under racial competence

Next, I examine voter welfare and the implications of the need for contextual knowledge on outcomes. Remark 4 depicts the impact of prior beliefs on voter welfare.

Remark 4. *Voter welfare satisfies:*

1. *Increasing the prior belief that the challenger is racially competent always increases welfare:*

$$\frac{\partial W}{\partial \pi_C} > 0.$$

2. *Increasing the prior belief that the incumbent is racially competent decreases welfare whenever*

$$\beta > (2 - \rho)\sigma^2.$$

Voter welfare increases in challenger quality and decreases in incumbent quality when rents from office are large (Bils 2023). This parallels previous results, but the addition of the ρ parameter modifies the conditions under which it applies: uncertainty about the need for contextual knowledge shifts the thresholds at which rents reduce welfare.

Bils (2023) engages with two implications of voter welfare on incumbent policymaking authority: when office benefits are low, policymaking should be made by the incumbent; otherwise, either the incumbent should be stripped of policymaking authority, or the voter

must always reelect the incumbent. The logic is that when reelection concerns are muted, incumbents are less inclined to distort policy. I qualify both implications by showing that the ρ parameter alters this tradeoff. Specifically, Implication 1 demonstrates that uncertainty about the need for racial competence both circumscribes the voter’s selection mechanism and lowers the rent threshold at which incumbents posture.

Implication 1. *Increasing the probability that the next period is universal lowers the β required for π_I to reduce voter welfare.*

As ρ rises, the office-benefit threshold falls, so incumbents posture at lower rent levels and welfare suffers. In effect, contextual knowledge renders the rent cutoff endogenous to the policy domain. When the probability of a universal policy is high, incumbents are more willing to posture, and voters are less able to rely solely on rents to discipline behavior. Racial competence, therefore, functions as a selection mechanism, but one whose efficacy depends on the interaction of ρ and β . By lowering the office benefit threshold, Implication 1 shows that higher ρ makes elections more fragile as a selection device.

Implication 1 highlights that the screening role of elections depends jointly on ρ and β . What remains is to consider how voter beliefs about incumbent competence interact with this tradeoff. Implication 2 turns to this issue, showing how changes in π_I alter welfare once contextual knowledge is taken into account.

Implication 2. *If office benefits are low, elections screen for racially competent incumbents and improve voter welfare. When office benefits are high, incumbents posture to appear competent, reducing welfare unless policymaking authority is constrained or contextual knowledge is more strictly required.*

This implication parallels the argument in Mansbridge (2009) that descriptive representation is best seen as a selection mechanism. Voters choose representatives not only for racial competence but also for their capacity to translate that competency into substantive policy. When office incentives are low, descriptive cues and the accountability mechanism improve voter welfare; however, when incentives are strong, incumbents are more willing to distort policy, undermining this mechanism and making elections less effective as a tool for racial screening. Voter welfare is shaped not only by policy outcomes but also by prior beliefs about the racial competence of incumbents and challengers. For Multiracial incumbents, racial ambiguity makes these priors especially salient. If voters view the incumbent as a coracial, they are more likely to interpret their actions as evidence of competence. In contrast, if they view the challenger as more racially competent, the incumbent faces a narrower scope for posturing. In this way, perceptions of racial competence act as a channel through which descriptive representation influences strategic behavior.

In the simplest case, one can interpret a voter’s prior belief about the challenger’s racial competence as their perception of the challenger’s co-racial status: a higher π signifies a Black politician, while a lower π implies a White challenger. From the incumbent’s perspective, these perceptions determine the attractiveness of mixing. A firmer belief in the

incumbent’s competence increases the credibility of their postures and makes mixing more valuable; conversely, a firmer belief in the challenger’s competence reduces the payoff to mimicry and constrains the uninformed type’s strategic options. Proposition 4 formalizes this relationship.

Proposition 4 (Impact of Prior Beliefs on Mixing Probabilities). *Let $\bar{\Sigma}$ and $\underline{\Sigma}$ denote the equilibrium mixing probabilities associated with right-posturing and left-posturing, respectively. Then,*

1. *An increase in the prior belief that the incumbent is racially competent raises the right-posturing mixing probability, $\frac{d\bar{\Sigma}}{d\pi_I} > 0$, and lowers the left-posturing mixing probability, $\frac{d\underline{\Sigma}}{d\pi_I} < 0$.*
2. *An increase in the prior belief that the challenger is racially competent lowers the right-posturing mixing probability, $\frac{d\bar{\Sigma}}{d\pi_C} < 0$, and raises the left-posturing mixing probability, $\frac{d\underline{\Sigma}}{d\pi_C} > 0$.*

By connecting voter beliefs to the incumbent’s actions, Proposition 4 illustrates how racial competence conditions both the direction and magnitude of mixing. The direct effect is that when voters are more confident that the incumbent is competent ex-ante (i.e., when the challenger is perceived as White), they are more likely to view the incumbent as racially aligned, thereby increasing the credibility of posturing. The indirect effect follows from the same belief. Stronger perceptions of racial competence widen the equilibrium range over which the incumbent can mix, expanding the set of posturing strategies consistent with voter indifference. By contrast, when voters believe the challenger is racially competent (i.e., the challenger is perceived as Black), the uninformed incumbent has less room to maneuver. Posturing carries a higher risk of exposure, and the safer strategy is to minimize losses by selecting the first-best outcome. The magnitude of this tradeoff is further shaped by contextual knowledge: as the probability of a universal domain rises, the thresholds that discipline posturing shift, making the incumbent’s willingness to mix less sensitive to perceptions of racial competence.

Substantively, this connects to the literature on reflected appraisals among Multiracial Americans. Identification is shaped less by self-perception than by how Monoracial peers evaluate mixed-race individuals (Khanna 2004, 2010; Sims 2016), and these evaluations are moderated by racial context (Rockquemore and Brunsma 2002a). The model captures this dynamic by treating voter beliefs about competence as priors that directly affect the incumbent’s willingness to mix. For Multiracial incumbents, being perceived as coracial heightens the pressure to act within group constraints, especially when racial competence is demanded. In equilibrium, this pressure manifests as a greater incentive to posture.

Extensions

While the baseline model provides a foundation for the role racial competence plays in structuring Multiracial behavior, there are further political considerations that shape how racial competence affects incumbents' behavior. Racial gerrymandering reshapes the electorate's capacity to condition on racial competence, generating behavioral and welfare consequences that vary with the issue environment. Finally, legislative constraints through caucus membership serve as insurance for Multiracial politicians, safeguarding against the uncertainty of future demands for racial competence.

Racial Gerrymandering: Cracking and Packing

A core application of my model involves racial gerrymandering. My model disentangles the bundled effect of gerrymandering on incumbent behavior, separating the ideological and racial competence dimensions.¹³ Rather than merely relocating the median voter, redrawing district lines determines whether the pivotal voter demands racial competence from the incumbent. This creates accountability for particularized policy rather than ideological position alone. This distinction is especially significant for Multiracial politicians, who may clearly align with a left-right ideological spectrum but, due to a lack of contextual knowledge, are unable to meet the particularized policy needs of their Black constituency. Taking a reduced-form approach, I formalize this through $\tau \in [0, 1]$, the probability that the pivotal voter does not condition their vote on racial competence. We can think about this in terms of racial gerrymandering—when $\tau = 1$, the district becomes cracked, and the need for racial competence becomes less central to winning reelection; conversely, when $\tau = 0$, the district is packed, resulting in voters conditioning on racial competence.¹⁴ This parameter thus captures the uncertainty gerrymandering introduces over which *type* of voter will be pivotal on election day.

This contribution fills a gap in existing scholarship on the impacts of racial gerrymandering on representation. Empirical scholars have questioned whether district composition improves substantive representation for Black voters (Cameron, Epstein, and O'halloran 1996; Canon 1999; Epstein and O'Halloran 1999; Lublin 1997; Swain 1995), focusing almost exclusively on the role gerrymandering plays in shifting the median voter's ideal point along an ideological dimension. The formal political economy literature mirrors this framing. The vast focus has been on studying the perverse effects of majority-minority mandates on partisan composition and policy outcomes (Shotts 2001, 2002).¹⁵ Across these contributions,

13. The results here are similar to an extension examining the role of ideology on incumbent behavior. See Section D of the Supplementary Materials.

14. The relationship between racial gerrymandering and representation can also be reinterpreted as targeted voting costs that shift the composition of voters within a district (Judd, Sasso, and Steel 2026).

15. Another strand of political economy models, optimal designer problems, asks how a partisan gerrymanderer allocates voters across districts to maximize seat share (Friedman and Holden 2008; Gul and Pesendorfer 2010; Kolotilin and Wolitzky 2026; Moscariello 2026). Yet the focus remains on how a partisan-

the operative assumption is that redistricting matters because it relocates the median voter along a left-right spectrum.¹⁶ What remains unexplored, and what my model brings into focus, is whether redistricting also changes the *dimension* along which incumbents are evaluated, and specifically, whether racial competence is among the criteria on which they can be held accountable at all.

Let there be a lottery over the district’s demographic makeup such that the incumbent is uncertain about τ . Here, the incumbent knows that the current period requires racial competence ($d_1 = P$) and that they are in a packed district ($\tau = 0$). But, there is uncertainty about whether racial gerrymandering in the next period will significantly alter the composition of their new district, affecting whether ρ is evaluated at the ballot box. While ρ is independent of τ , the incumbent still factors in this uncertainty about whether the issue of the day will require racial competence. The new incentive compatibility constraint for the uninformed type is

$$\underbrace{\beta - \sigma^2}_{\text{Equilibrium Payoff}} - \underbrace{[1 - \pi_C]\sigma^2}_{V(C)} = \underbrace{-x^2}_{\text{Policy Indifference}} + \underbrace{2\beta - (2 - \rho - \tau - \rho\tau)\sigma^2}_{\text{Total Expected Payoff}}. \quad (3)$$

Solving results in the policy bounds $\bar{x}^P(\rho, \tau)$ and $\underline{x}^P(\rho, \tau)$ where the uninformed incumbent is indifferent: $\bar{x}^P(\rho, \tau) = \sqrt{\beta - (\pi_C - \rho - \tau - \rho\tau)\sigma^2}$. This leads to the behavioral effect of racial gerrymandering on the incumbent:

Proposition 5 (Behavioral Effect of Racial Gerrymandering). *Suppose the district faces redistricting uncertainty $\tau \in [0, 1]$. Then,*

1. **Policy Extremity.** *An increase in the probability that the district is cracked leads to an increase in the extremity of the policy choice:*

$$\frac{\partial \bar{x}^P(\tau)}{\partial \tau} > 0 \quad \text{and} \quad \frac{\partial \underline{x}^P(\tau)}{\partial \tau} < 0.$$

2. **Winning Probability.** *As τ increases, the winning probability for the uninformed incumbent increases:*

$$\frac{\partial P_{win}^L}{\partial \tau} > 0.$$

3. **Interaction effect on the state threshold.** *The marginal benefit of cracking on the state threshold increases as ρ increases:*

$$\frac{\partial^2 \omega^P(\rho, \tau)}{\partial \tau \partial \rho} > 0,$$

when β is sufficiently large.

leaning designer can maximize their share of the vote within a particular district.

16. Groll and O’Halloran (2024) do move beyond analyzing ideology by focusing on the electoral process as a game where a district’s racial composition determines the allocation of distributive benefits. However, their model contains no accountability mechanism through which district composition alters incumbent behavior.

4. **Interaction effect on the winning probability.** *The marginal benefit of cracking on the incumbent’s winning probability increases as ρ increases:*

$$\frac{\partial^2 P_{win}^L}{\partial \tau \partial \rho} > 0,$$

when β is sufficiently large.

Proposition 5 resolves a core tension in the gerrymandering-electoral effects literature. In particular, Cameron, Epstein, and O’halloran (1996) find no behavioral effect of gerrymandering on politicians, arguing that redistricting “does not change the voting pattern of any given representative.” Yet, Lublin (1999) highlights that their hypothesis is unable to separate district composition from partisanship. I show that once you introduce a competency channel distinct from partisanship, there is a behavioral effect: first, cracking increases expected heterogeneity in the next period; second, under heterogeneity, the uninformed type’s optimal strategy shifts toward extremity because the competency accountability mechanism is diluted. This behavioral channel is distinct from both the ideological mechanism in Cameron, Epstein, and O’halloran (1996)’s framework and the selection channel in Canon (1999), who shows that district composition shapes which types of representatives emerge rather than how a given incumbent makes policy decisions.

Moreover, while cracking promotes voter heterogeneity, the marginal effect on policy extremity is amplified by ρ when office rents are large relative to next-period uncertainty.¹⁷ When this condition holds, as the probability that the next period is universal increases, cracking’s protective value for the uninformed incumbent grows. As a result, cracking operates most powerfully when the issue environment is universal in nature, and voter scrutiny on racial competence is weakest.

Implication 3. *For Multiracial incumbents, the behavioral effect of racial gerrymandering scales with office benefits: cracking is most consequential when the benefits of office are large relative to the screening risk of expanding the policy region.*

Implication 3 shows that racial gerrymandering is most consequential when office rents are high compared to the risk of policy extremity. A significant consequence is that individuals in charge of redistricting face increasing returns to cracking as the political environment within a district becomes more universal. Even though a designer may make a district more racially heterogeneous (for example, to increase substantive representation at the aggregate level à la Cameron, Epstein, and O’halloran (1996) and Groll and O’Halloran (2024)), cracking does its sharpest behavioral work precisely where the issue of the day lacks the need for racial competence: with no countervailing voter accountability via racial competence to discipline the incumbent, the uninformed type expands the policy region to secure reelection when office benefits are high. This refines the “sometimes pack but never crack”

17. If $\beta > \frac{(1+2\pi_C-\rho-\tau-\rho\tau)\sigma^2}{2}$, office rents are large enough that the incumbent absorbs the screening risk from extending the pooling region, and cracking’s marginal behavioral effect grows with ρ . If office benefits are low, the screening risk dominates, and the behavioral effect of cracking with respect to ρ attenuates.

argument in Friedman and Holden (2008) by identifying a behavioral channel absent from their framework.¹⁸

Voter Welfare under Racial Gerrymandering

Turning now to voter welfare, Proposition 6 shows two important results.

Proposition 6 (Voter Welfare under Racial Gerrymandering). *Voter welfare satisfies:*

1. *Welfare is decreasing in the probability of cracking:*

$$\frac{\partial W}{\partial \tau} < 0.$$

2. *The marginal effect of cracking on welfare is decreasing in ρ :*

$$\frac{\partial^2 W}{\partial \tau \partial \rho} < 0.$$

First, voter welfare is decreasing in τ . By splintering voters who care about racial competence, cracking circumscribes their ability to hold the uninformed incumbent accountable for distorting policy. Part 2 amplifies this baseline result, with three implications for racial gerrymandering. First, ρ generates a welfare-accountability complementarity—as the issue environment universalizes, voter scrutiny on racial competence recedes, and the welfare cost of cracking is most acute. While there may be no perverse effects along the ideological dimension (Shotts 2001, 2002), gerrymandering harms racial-group accountability precisely where one might expect those harms to vanish.¹⁹ Second, packing serves voters not only by increasing descriptive representation but, when the issue of the day does not require contextual knowledge, by preserving the competency-accountability mechanism that cracking would otherwise dilute. Finally, the welfare loss from cracking grows as the political environment universalizes, making the normative case against racial gerrymandering itself endogenous to the issue environment. This finding speaks directly to debates over the continued necessity of Section 2 of the Voting Rights Act.

18. Moreover, it highlights an unexplored consequence of cracking not found in the broader optimal designer literature. Whereas existing models focus on how partisans allocate voters across districts to maximize seat share (Gul and Pesendorfer 2010; Kolotilin and Wolitzky 2026), when voters are unconcerned with racial competence, cracking allows the uninformed incumbent to extend policy beyond the state threshold without facing electoral discipline.

19. This finding is not obvious in the formal literature on gerrymandering, where there is no mechanism for voters to evaluate legislator competency. Distributing minority voters across districts may improve aggregate substantive representation (Cameron, Epstein, and O’halloran 1996; Groll and O’Halloran 2024), but at the cost of within-district minority voter welfare, with that cost largest when the issue environment makes racial competence least electorally visible.

Legislative Constraints

A final avenue of increasing perceptions of racial competence is joining social identity-oriented caucuses in the legislature. Canon (1999) shows that identity caucuses serve a primary purpose of constraining legislative behavior, providing resources to coracial constituencies, and ensuring powerful leadership positions for members. Rendleman (2023), focusing on the constraining mechanism of caucuses, highlights that the primary function of institutional organizations is to act as social monitors, pressuring astray members from potentially deviating from coracial goals.²⁰ Here, I focus on the role identity caucuses have on constraining Multiracial incumbents' ability to distort policy and the influence racial competence has on their willingness to constrain.

Assume that there is some legislative constraint $[-\Psi, \Psi]$ such that $\Psi < \bar{x}^P(\rho)$.²¹ An analogous case is supported for $-\Psi > \underline{x}^P(\rho)$. Moreover, let $V_C(\Psi)$ be the voter's continuation value for electing the challenger given the constraint. Up to this point, I have assumed that $d_1 = P$. However, to illustrate a conservative aspect of caucus membership, I assume now that $d_1 = U$. The argument I'm going to make is that caucus membership serves as a signal of racial competence through the channel ρ operates, voters' perceptions. A universal first period provides the most conservative test of this racial competence's role, as voters do not require contextual knowledge *ex ante*. Multiracial incumbents, therefore, are less likely to engage in caucus participation *a priori*. Yet, Lemma 3 shows that this is not the case.

Lemma 3 (Symmetric Legislative Constraint). *Suppose the legislative constraints are strong ($\Psi < \bar{x}^P(\rho)$).*

1. *Assume the incumbent is popular ($\pi_I > \pi_C$).*

- *Informed Incumbent: If $\omega \geq 0$ then the incumbent chooses $x_1 = \Psi$. If $\omega < 0$, then the incumbent chooses $x_1 = -\Psi$.*
- *Uninformed Incumbent: The incumbent chooses $x_1 = \Psi$ with probability $1 - F(0)$ and $x = -\Psi$ with probability $F(0)$.*
- *Voter: The voter always reelects the incumbent.*

2. *Assume the incumbent is unpopular ($\pi_I < \pi_C$).*

- *Informed Incumbent: If $\omega \geq 0$ then the incumbent chooses $x_1 = \Psi$. If $\omega < 0$, then the incumbent chooses $x_1 = -\Psi$.*
- *Uninformed Incumbent: The incumbent chooses $x_1 = \Psi$ with probability $\frac{\pi_I(1-\pi_C)}{\pi_C(1-\pi_I)}(1 - F(0))$ and $x = -\Psi$ with probability $\frac{\pi_I(1-\pi_C)}{\pi_C(1-\pi_I)}F(0)$.*
- *Voter: The voter reelects the incumbent with probability $\mu(\Psi) = \frac{\Psi^2 + \sigma^2}{\beta - (1-\rho)\sigma^2 - V_C(\Psi)}$.*

20. Lemi (2018) finds the caucus participation for Multiracial politicians is difficult in large part due to this social monitoring mechanism. Because they doubt the racial competence of the Multiracial, caucuses are less likely to allow them to participate.

21. When constraints are weak, the incumbent distorts policy as characterized in Proposition 1.

Similar to Bills (2023), when the incumbent is popular, caucus constraints diminish voters’ accountability mechanism, since both types of incumbents pool on the constraint and are always reelected. By contrast, if the incumbent is unpopular, the informed type still selects the constraint in equilibrium. Still, the uninformed type requires an incentive to do so, rather than choosing their optimal payoff, $x_1 = 0$, given that the challenger is ex-ante more likely to be racially competent. This incentive arises when voters mix between reelecting the incumbent and supporting the challenger, making the incumbent indifferent. Driving the voter’s reelection probability are perceptions of future racial competence, both indirectly via their continuation value and directly in the probability itself. Perceptions drive Multiracial behavior (Khanna 2004, 2010; Sims 2016), and uncertainty about racial competence increases their willingness to engage in potentially risky political action such as polarization (Davenport 2016; Davenport, Franco, and Iyengar 2022).²² Therefore, uncertainty raises the salience for uninformed incumbents to constrain themselves *even if the present environment doesn’t necessitate it*. In turn, the incumbent mixes between setting $x_1 = \Psi$ with some probability and $x_1 = 0$, which then keeps voters indifferent between retaining the incumbent and electing the challenger.

Voters’ perceptions and incumbents’ uncertainty about ρ drive Multiracial behavior. Proposition 7 highlights how racial competence mediates the effect of the caucus constraint on the voter’s reelection probability.

Proposition 7 (Impact of ρ on Voters’ Perceptions). *As ρ increases, the reelection probability for the uninformed incumbent increases.*

The results of Proposition 7 indicate that constraints improve the incumbent’s reelection probability, and that ρ mediates this effect. By constraining themselves to the present universal period, incumbents guard against the possibility that an electorate requiring racial competence will arise tomorrow. In this sense, the constraint functions as a coracial commitment: a costly signal of racial competence even when the incumbent is not inherently competent. For Multiracial incumbents, joining the caucus—even when the immediate policy domain does not require racial competence—enhances the perception that they are racially competent should the next period demand it. By improving one’s perceptions, caucus membership also serves as another important mechanism of reflected appraisal. Implication 4 provides evidence of this fact.

Implication 4. *For Multiracial incumbents, membership in a racial caucus is a mechanism of reflected appraisal: by joining a racial caucus, uninformed types reduce the burden of proving racial competence.*

Although the comparative statics suggest that reelection prospects are higher when tomorrow is universal, this effect arises precisely because the incumbent constrained themselves today rather than pursuing their unconstrained policy optimum. The insurance is thus

22. Multiracial behavior has also been framed around risk-taking action to appeal to their Monoracial Black peers, as Fryer et al. (2012) empirically shows.

forward-looking: present discipline secures future electoral viability.²³ This also provides a new channel for why incumbents whose values do not align with the caucus otherwise conform to the constraints imposed. Rendleman (2023) shows that caucus membership serves as a racialized social constraint (White and Laird 2020), sanctioning members who deviate from their legislative priorities. The threat of needing to be racially competent in the future also acts as an indirect sanction, not through legislative peers but through voters’ perceptions. Multiracials worry about their caucus perceptions (Lemi 2018), but they are also driven by voters’ expectations within the electoral context. Thus, caucus membership provides both institutional (Canon 1999) and electoral utility.

Discussion and Conclusion

In this paper, I have introduced the concept of domain-specific competency and applied it to a paramount issue in race and ethnic politics: where an emerging group of Multiracial politicians fits into the descriptive versus substantive representation debate. In doing so, my model demonstrates that racial competence conditions the behavior of multiracial officeholders through a feedback dynamic that is driven by voters’ perceptions. Uninformed politicians lack racial competence but desire to be perceived as such. To credibly signal to voters that they are racially competent, uninformed politicians exaggerate policy. Significantly, uncertainty about racial competence affects both the extent of posturing and incumbents’ willingness to posture in equilibrium. Because incumbents are more willing to posture when they are perceived as racially competent *ex ante*, elections serve as a poor screening mechanism unless constraints are enforced or voters require contextual knowledge in following periods. Introducing voter heterogeneity via racial gerrymandering creates behavioral incentives to expand policy extremity to win reelection, with significant impacts on voter welfare. Finally, caucus membership effectively constrains policy distortion, but only through the channel of uncertainty about the future need for racial competence.

My model shows that political alignment does not equate to racial competence. Multiracials are more likely to align with their monoracial minority peers (Davenport, Iyengar, and Westwood 2022). Such alignment often translates into more ideologically extreme positions (Davenport 2016; Davenport, Franco, and Iyengar 2022). Leslie and Sears (2022) shows that for Black–White Multiracials in particular, the legacy of hypodescent structures political alignment, offering a precise mechanism for why Multiracials are aligned with their minority background. My model highlights how the desire to be perceived as racially competent underlies much of the literature on alignment. Multiracials polarize when uncertainty about the need for racial competence is high. They also participate in racial caucuses to mitigate the risk of being perceived as unaligned in the future. In this way, my model provides a microfoundation for how Multiracials can be politically aligned yet still lack racial

23. Section C.2.2 of the Supplementary Materials shows the analogous comparative static for $d_1 = P$. While the result is similar, reelection probabilities have a steeper slope when $d_1 = U$ due to the positive σ^2 term in the voter’s reelection function.

competence.

Moreover, the model shows that racial competence is a significant constraint on Multiracial behavior, structuring their actions even when voters face information barriers. In standard models of democratic theory, voters *authorize* politicians to represent them and hold them *accountable* through elections (Pitkin 1967; Young 1997, 2002). Racial competence facilitates representation for disadvantaged groups by acting as a form of insurance, ensuring that the politician will act on behalf of the marginalized group’s interest if the deliberative body is unaware of, or unwilling to pursue, those interests (Dovi 2002; Mansbridge 1999, 2009; Williams 1993). What has remained unclear, however, is whether this logic of racial competence extends to Multiracial politicians and their constituencies. Here, I show that though Multiracial politicians have incentives to posture, racial competence remains a tool for ensuring minority voters receive credible representation.

Building on this, I qualify Bils (2023) by showing that racial competence functions as a selection mechanism by raising the level of office rents required to lower voter welfare. Within the political economy literature, voter welfare decreases as prior beliefs about an incumbent’s competency increase (Bils 2023; Canes-Wrone, Herron, and Shotts 2001). Incumbents are more willing to posture when they are perceived as competent, which leads to worse outcomes for voters (Bils 2023). The findings presented here indicate that this willingness is influenced by uncertainty regarding the necessity of racial competence. By increasing the required office rents, racial competence improves voters’ ability to distinguish between competent and incompetent types in equilibrium.

Beyond the application to racial politics in the U.S., this paper has broader implications for theories of accountability by introducing domain-specific competency. Previously, competency was viewed as an exogenous source of variation in an incumbent’s behavioral calculus (Bils 2023; Canes-Wrone and Shotts 2007; Fox 2007; Maskin and Tirole 2004). By rendering competency endogenous to the policy space, voters’ perceptions of an incumbent’s ability to meet the unique needs of their constituency now drive decision-making. This general framework extends to other questions related to representation and policymaking: how women legislators navigate “women’s issues” when cross-pressured by the need to present a masculine identity (Atkinson and Windett 2019), how working-class politicians approach economic policy in legislatures whose membership is concentrated among the wealthy (Carnes 2013), how veterans handle issues related to defense (Lupton 2017, 2022), and finally, how LGBTQ officeholders address LGBTQ rights (Reynolds 2013). Future research should engage more critically with domain-specific competency when addressing group-based representation.

Future research should also empirically test the model’s predictions on Multiracial respondents. In particular, the ideological arguments in Section D of the Supplementary Materials are closely tied to the literature on racial attitudes and Multiracial political behavior (Davenport 2016; Davenport, Franco, and Iyengar 2022). While Davenport (2016) and Davenport, Franco, and Iyengar (2022) demonstrate that psychological dispositions lead Multiracial respondents to be more liberal than their monoracial peers, my model provides a mechanism for evaluating the strategic underpinnings of this behavior. Relatedly, there

are avenues to investigate the impact of racial gerrymandering on Multiracial representation, specifically by empirically testing the core result that redistricting has a behavioral impact on incumbent behavior. Ultimately, the model provides a generative framework for examining the formal extensions and empirical implications of Multiracial behavior.

With the recent Supreme Court ruling in *Louisiana v. Callais*, the implications of racial gerrymandering for incumbent behavior and voter welfare become significant. Speaking for the majority, Justice Samuel Alito argued that “in short, §2 imposes liability only when the evidence supports a strong inference that the State *intentionally* drew its districts to afford minority voters less opportunity because of their race” (*Louisiana v. Callais* 2026, 26, emphasis mine).²⁴ By shifting the evidentiary burden from effects to intent, *Callais* relaxes the §2 constraint that previously limited designers’ ability to crack minority districts without legal consequence. Proving discriminatory intent is a more difficult standard to meet, due in part to the conflation of partisan and racial motivations in former Confederate states (Chemerinsky 2026).²⁵ And while the debate over *Callais* has centered on whether §2 guards against discriminatory effects or only discriminatory intent, this paper suggests that the question is incomplete. At stake is not only who gets elected, but whether voters retain the ability to discipline the representatives they elect.²⁶ Racial gerrymandering does not merely dilute the vote; it severs the feedback between constituents and incumbents on which representation depends. Multiracial politicians, whose competency is most contested and most contingent on the district they serve, will feel that severance first.

24. While *Callais* does not formally overrule *Thornburg v. Gingles*, its reinterpretation of §2 to require intentional discrimination substantially constrains the evidentiary path by which plaintiffs can establish a *Gingles* claim.

25. Moreover, as Chemerinsky (2026) notes, the groundwork for hollowing out Section 2 of the Voting Rights Act was laid by a series of decisions that eroded judicial deference to Congressional authority in the lead up to *Callais*.

26. The model predicts that more cracked districts will push Multiracial incumbents toward more extreme policy and erode the accountability mechanism that ensures racial competence, at a substantial cost to voter welfare (Propositions 5 and 6).

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Supplementary Materials for

Jack of All Trades, Master of Some:

Racial Competency and the Limits of Multiracial Representation

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Contains proofs of all results in the main text, together with an extension to ideological bias.

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A Proofs of Main Findings

A.1 Second-period Behavior

Lemma 1 (Impact of ρ on Continuation Values). *The voter's continuation value from electing the challenger is strictly increasing in ρ , and converges to 0 as ρ approaches 1.*

Proof of Lemma 1. let

$$V(C) = -[1 - \pi_C](1 - \rho)\sigma^2.$$

To compute the partial derivative of $V(C)$ with respect to ρ :

$$\frac{\partial V(C)}{\partial \rho} = -[1 - \pi_C]\sigma^2 \cdot \frac{\partial}{\partial \rho}(1 - \rho).$$

Since

$$\frac{\partial}{\partial \rho}(1 - \rho) = -1,$$

It follows that

$$\frac{\partial V(C)}{\partial \rho} = -[1 - \pi_C]\sigma^2 \cdot (-1) = [1 - \pi_C]\sigma^2.$$

Because $\pi_C \in (0, 1)$ implies that $1 - \pi_C > 0$ and since $\sigma^2 > 0$, we have

$$\frac{\partial V(C)}{\partial \rho} > 0.$$

Thus, $V(C)$ is strictly increasing in ρ .

Next, consider the limit as ρ approaches 1:

$$\lim_{\rho \rightarrow 1} V(C) = \lim_{\rho \rightarrow 1} (-[1 - \pi_C](1 - \rho)\sigma^2).$$

Since

$$\lim_{\rho \rightarrow 1} (1 - \rho) = 0,$$

we obtain

$$\lim_{\rho \rightarrow 1} V(C) = -[1 - \pi_C]\sigma^2 \cdot 0 = 0.$$

Hence, as ρ approaches 1, the continuation value $V(C)$ converges to 0. □

Lemma 2 (Voter Decision Rule). *After observing first-period policy x_1 in equilibrium, the voter (1) reelects the incumbent if $\tilde{\pi}(x_1) > \pi_C$, (2) reelects the challenger if $\tilde{\pi}(x_1) < \pi_C$, and (3) is indifferent if $\tilde{\pi}(x_1) = \pi_C$.*

Proof of Lemma 2. Define the voter's payoff difference as

$$\Delta(x_1) = V(I | x_1) - V(C)$$

We want to show that $\Delta(x_1) > 0$ if and only if $\tilde{\pi}(x_1) > \pi_C$.

Let

$$V(I | x_1) = -[1 - \tilde{\pi}(x_1)](1 - \rho)\sigma^2, \quad V(C) = -[1 - \pi_C](1 - \rho)\sigma^2.$$

Then,

$$\Delta(x_1) = -[1 - \tilde{\pi}(x_1)](1 - \rho)\sigma^2 - \{-[1 - \pi_C](1 - \rho)\sigma^2\}.$$

Combining like terms results in,

$$\Delta(x_1) = (1 - \rho)\sigma^2\{[1 - \pi_C] - [1 - \tilde{\pi}(x_1)]\}.$$

Simplifying further,

$$\Delta(x_1) = (1 - \rho)\sigma^2[\tilde{\pi}(x_1) - \pi_C].$$

Since $(1 - \rho)\sigma^2 > 0$, the sign of $\Delta(x_1)$ is entirely determined by $[\tilde{\pi}(x_1) - \pi_C]$. Therefore, $\Delta(x_1) > 0 \iff \tilde{\pi}(x_1) > \pi_C$.

An analogous case can be shown for $\Delta(x_1) < 0 \iff \tilde{\pi}(x_1) < \pi_C$ and $\Delta(x_1) = 0 \iff \tilde{\pi}(x_1) = \pi_C$. $\square$

A.2 Complete Information Benchmark

Remark 1. *Under the complete information benchmark, there is a subgame perfect Nash equilibrium in which the high-type incumbent always selects $x_1 = \omega_1$, and the low-type incumbent selects*

$$x_1 = \begin{cases} \omega_1, & \text{if } d_1 = U, \\ 0, & \text{if } d_1 = P. \end{cases}$$

The voter always reelects the high-type and rejects the low type.

Proof of Remark 1. It is straightforward to show that both types of incumbents never deviate when $d_1 = U$. Thus, I focus on the uninformed incumbent when $d_1 = P$. In this case, suppose that the incentive compatibility constraint is given by

$$\underbrace{\beta - \sigma^2}_{\text{Equilibrium Payoff}} - \underbrace{[1 - \pi_C]\sigma^2}_{V(C)} \geq \underbrace{2\beta - (2 - \rho)\sigma^2}_{\text{Deviation Payoff}}.$$

We begin by writing the inequality as

$$(\beta - \sigma^2) - [1 - \pi_C]\sigma^2 \geq 2\beta - (2 - \rho)\sigma^2.$$

Subtracting $\beta - \sigma^2$ from both sides gives

$$-[1 - \pi_C]\sigma^2 \geq [2\beta - (2 - \rho)\sigma^2] - (\beta - \sigma^2).$$

Simplify the right-hand side:

$$\begin{aligned} 2\beta - (2 - \rho)\sigma^2 - (\beta - \sigma^2) &= 2\beta - (2 - \rho)\sigma^2 - \beta + \sigma^2 \\ &= \beta - [(2 - \rho)\sigma^2 - \sigma^2] \\ &= \beta - [(2 - \rho - 1)\sigma^2] \\ &= \beta - (1 - \rho)\sigma^2. \end{aligned}$$

Thus, the inequality becomes

$$-[1 - \pi_C]\sigma^2 \geq \beta - (1 - \rho)\sigma^2.$$

Multiplying both sides by -1 (which reverses the inequality) yields

$$[1 - \pi_C]\sigma^2 \leq (1 - \rho)\sigma^2 - \beta.$$

Adding β to both sides,

$$[1 - \pi_C]\sigma^2 + \beta \leq (1 - \rho)\sigma^2,$$

and then subtracting $[1 - \pi_C]\sigma^2$ from both sides gives

$$\beta \leq (1 - \rho)\sigma^2 - [1 - \pi_C]\sigma^2.$$

Factoring σ^2 on the right-hand side,

$$\beta \leq [(1 - \rho) - (1 - \pi_C)]\sigma^2.$$

Since

$$(1 - \rho) - (1 - \pi_C) = \pi_C - \rho,$$

we obtain

$$\beta \leq (\pi_C - \rho)\sigma^2,$$

or equivalently,

$$\pi_C \sigma^2 - \rho \sigma^2 \geq \beta.$$

Under the assumption that $\pi_C \in (0, 1)$ and $\sigma^2 > 0$, the right-hand side $\pi_C \sigma^2 - \rho \sigma^2$ is strictly positive for any $\rho < \pi_C$. Hence, there exists a range of β satisfying

$$\beta \leq (\pi_C - \rho) \sigma^2.$$

Therefore, when $d_1 = P$, the uninformed incumbent has no incentive to deviate from their equilibrium strategy. $\square$

A.3 Uncertainty about Racial Competency

A.3.1 Low-value Office

Remark 2. *If office rents are low so that $\beta \leq \pi_C \sigma^2 - \rho \sigma^2$, then there exists a perfect Bayesian equilibrium in which the high-type incumbent selects $x_1 = \omega_1$ and the low-type incumbent selects $x_1 = 0$.*

Proof of Remark 2. Assume that there exists a perfect Bayesian equilibrium when office rents are low. In such an equilibrium, the informed incumbent chooses

$$x_1 = \omega_1,$$

while the uninformed incumbent chooses

$$x_1 = 0.$$

By weak consistency, when a voter observes $x_1 = \omega_1$, she assigns probability 1 to the incumbent being high-type. Moreover, by sequential rationality, the voter's updated belief, $\tilde{\pi}_I(x_1)$, satisfies

$$\tilde{\pi}_I(x_1) > \pi_C,$$

ensuring that re-election occurs only when the incumbent is perceived as an informed type.

To complete the argument, we must show that the uninformed type has no profitable deviation to mimic the informed type. Under this condition, the incentive compatibility constraint simplifies to:

$$\pi_C \sigma^2 \geq \beta.$$

This inequality holds for β sufficiently small, ensuring that the low-type's deviation is not profitable. Thus, a separating, first-best equilibrium exists when the policy domain is particular. $\square$

A.3.2 High-value Office

Proposition 1 (Perfect Bayesian Equilibria with High Office Rents). *The set of Perfect Bayesian Equilibria of the model is characterized as follows:*

1. Voter strategy.

- If $x_1 \in (\underline{x}^P(\rho), \bar{x}^P(\rho))$, the voter rejects the incumbent.
- If $x_1 \leq \underline{x}^P(\rho)$ or $x_1 \geq \bar{x}^P(\rho)$, the voter re-elects the incumbent.

2. Informed incumbent.

- If $\omega_1 \in [0, \omega^P(\rho))$, the informed type overreacts by setting $x_1 = \bar{x}^P(\rho)$.
- If $\omega_1 \in (-\omega^P(\rho), 0]$, the informed type overreacts by setting $x_1 = \underline{x}^P(\rho)$.
- If $|\omega_1| > \omega^P(\rho)$, the informed type plays the first-best by choosing $x_1 = \omega_1$.

3. Uninformed incumbent.

- With probability $\bar{\sigma} \in [0, \min\{1, \bar{\Sigma}\}]$, the uninformed type postures right by choosing $x_1 = \bar{x}^P(\rho)$.
- With probability $\underline{\sigma} \in [0, \min\{1 - \bar{\sigma}, \underline{\Sigma}\}]$, the uninformed type postures left by choosing $x_1 = \underline{x}^P(\rho)$.
- With residual probability $1 - \underline{\sigma} - \bar{\sigma}$, the uninformed type plays the first-best by setting $x_1 = 0$.

Off the path of play, assume the voter updates their belief with probability 1 that the incumbent is uninformed.

Proof of Proposition 1.

Uninformed Incumbent. The uninformed incumbent's utility is

$$-x^2 - \sigma^2.$$

Choosing any $x_1 \in (\underline{x}^P(\rho), \bar{x}^P(\rho)) \setminus \{0\}$ yields a payoff that is weakly worse than choosing $x_1 = 0$. Moreover, selecting any $x_1 > \bar{x}^P(\rho)$ or $x_1 < \underline{x}^P(\rho)$ (and getting re-elected) is strictly worse than selecting $x_1 = \bar{x}^P(\rho)$ or $x_1 = \underline{x}^P(\rho)$. Thus, both $\underline{x}^P(\rho)$ and $\bar{x}^P(\rho)$ render the uninformed type indifferent among the three actions:

$$x_1 = \bar{x}^P(\rho), \quad x_1 = \underline{x}^P(\rho), \quad \text{and} \quad x_1 = 0.$$

Consequently, the uninformed incumbent mixes over these three actions.

Informed Incumbent. For the informed incumbent, if $\omega_1 \geq \bar{x}^P(\rho)$ or $\omega_1 \leq \underline{x}^P(\rho)$, the optimal policy is to choose $x_1 = \omega_1$. Now, partition the state space such that for $\omega_1 \in [0, \bar{x}^P(\rho))$ the optimal policy to secure re-election is $x_1 = \bar{x}^P(\rho)$. If the informed type were to choose their bliss point $x_1 = \omega_1$, they would be removed from office. Thus, their payoffs are:

$$\text{If removed: } \beta - (1 - \pi_C)\sigma^2, \quad \text{if re-elected: } -(\bar{x}^P(\rho) - \omega_1)^2 + 2\beta.$$

The distortion cost $(\bar{x}^P(\rho) - \omega_1)^2$ is decreasing in ω_1 since

$$\frac{\partial [(\bar{x}^P(\rho) - \omega_1)^2 + 2\beta]}{\partial \omega_1} = -2(\bar{x}^P(\rho) - \omega_1) < 0.$$

Hence, if the type with $\omega_1 = 0$ prefers $x_1 = \bar{x}^P(\rho)$, then every $\omega_1 \in (0, \bar{x}^P(\rho))$ also prefers $x_1 = \bar{x}^P(\rho)$. To verify this, consider:

$$\begin{aligned} -x^2 + 2\beta &> \beta - (1 - \pi_C)\sigma^2, \\ -x^2 + \beta &> -(1 - \pi_C)\sigma^2, \\ (1 - \pi_C)\sigma^2 + \beta &> x^2, \\ (1 - \pi_C)\sigma^2 + \beta &> \beta - \pi_C\sigma^2 + \rho\sigma^2, \\ \sigma^2 - \pi_C\sigma^2 &> \pi_C\sigma^2 + \rho\sigma^2, \\ \sigma^2 &> \rho\sigma^2 \end{aligned}$$

which holds for all $\rho \in [0, 1)$. A similar argument applies for $\omega_1 \in (\underline{x}^P(\rho), 0)$ when choosing $x_1 = \underline{x}^P(\rho)$. Thus, the $\omega_1 = 0$ type prefers to select $x_1 = \bar{x}^P(\rho)$ (and be re-elected) over the bliss point $x_1 = 0$.

Voter Decision. A voter re-elects the incumbent if $x_1 \geq \bar{x}^P(\rho)$ or $x_1 \leq \underline{x}^P(\rho)$, and elects the challenger if $x_1 \in (\underline{x}^P(\rho), \bar{x}^P(\rho))$. By D1, any policy $x_1 \in (\underline{x}^P(\rho), \bar{x}^P(\rho)) \setminus \{0\}$ is off the equilibrium path, so that the voter's posterior satisfies $\tilde{\pi}_I(x_1) \leq \pi_C$ and the challenger is elected. We set $\tilde{\pi}_I(x_1) = 0$ for such x_1 . Since the uninformed type selects $x_1 = 0$, the voter's posterior is $\tilde{\pi}_I(0) = 0 < \pi_C$. In contrast, only the informed type selects $x_1 > \bar{x}^P(\rho)$ or $x_1 < \underline{x}^P(\rho)$; in those cases, $\tilde{\pi}_I(x_1) = 1 > \pi_C$. Finally, when $x_1 = \bar{x}^P(\rho)$, it is required that

$$\tilde{\pi}_I(x_1) \geq \pi_C,$$

so that the voter re-elects the incumbent. By Lemma 2, we have

$$\tilde{\pi}_I(x_1) = \frac{(F(\omega^P(\rho)) - F(0)) \pi_I}{\pi_I(F(\omega^P(\rho)) - F(0)) + (1 - \pi_C) \bar{\Sigma}} \geq \pi_C$$

which can be rearranged to

$$\frac{\pi_I(1 - \pi_C)}{\pi_C(1 - \pi_I)} (F(\omega^P(\rho)) - F(0)) \geq \bar{\Sigma}.$$

A similar argument holds for $x_1 = \underline{x}^P(\rho)$. □

Remark 3. *The equilibria identified in Proposition 1 survive D1.*

Proof of Remark 3.

Step 1: Sending off-the-path Message $\tilde{x}^P(\rho)$.

An arbitrary incumbent's type is given by $\tau \in \mathbb{R} \cup \{\phi\}$. Define $D_\sigma(\tau, x)$ as the set of mixed best responses (MBR) where the incumbent is strictly better off deviating to a new policy x and getting re-elected with probability μ versus following equilibrium x in a PBE σ . Let $D_\sigma^o(\tau, x)$ be the set of strategies that make τ indifferent. If $\tilde{x}^P(\rho)$ is off-path, D1 requires putting probability 0 on τ -type if there exist some τ' -type such that

$$D_\sigma(\tau, \tilde{x}^P(\rho)) \cup D_\sigma^o(\tau, \tilde{x}^P(\rho)) \subseteq D_\sigma(\tau', \tilde{x}^P(\rho)).$$

Setting $\omega_1 \geq \bar{x}^P(\rho)$ and $\omega_1 \leq \underline{x}^P(\rho)$, the informed incumbent never faces an incentive to deviate since they receive their highest payoff.

Examining the $\omega_1 \in [0, \bar{x}^P(\rho)]$ case for the Informed incumbent. The informed type selects $x_1 = \bar{x}^P(\rho)$ and their payoff

$$-(\bar{x}^P(\rho) - \omega_1)^2 + 2\beta.$$

If they deviate to some $x_1 = \tilde{x}^P(\rho)$, their payoff given $\mu(\tilde{x}^P(\rho))$ is

$$-(\tilde{x}^P(\rho) - \omega_1)^2 + \beta + \mu(\tilde{x}^P(\rho))\beta - (1 - \mu(\tilde{x}^P(\rho)))[(1 - \pi_C)\sigma^2].$$

Therefore, the incentive compatibility constraint is

$$\begin{aligned} & -(\tilde{x}^P(\rho) - \omega_1) + \beta + \mu(\tilde{x}^P(\rho))\beta - (1 - \mu(\tilde{x}^P(\rho)))[(1 - \pi_C)\sigma^2] \\ & > -(\bar{x}^P(\rho) - \omega_1) + 2\beta \\ \iff & \mu(\tilde{x}^P(\rho)) > \frac{(\tilde{x}^P(\rho) - \omega_1) - (\bar{x}^P(\rho) - \omega_1) + \beta + (1 - \pi_C)\sigma^2}{\beta + (1 - \pi_C)\sigma^2}. \end{aligned}$$

Solving the first-order condition results in

$$\begin{aligned}
& \frac{\partial(\tilde{x}^P(\rho) - \omega_1)^2}{\partial\omega_1} = 2(\tilde{x}^P(\rho) - \omega_1)(-1) \\
& \frac{\partial(\bar{x}^P(\rho) - \omega_1)^2}{\partial\omega_1} = 2(\bar{x}^P(\rho) - \omega_1)(-1) \\
\Longleftrightarrow & \frac{-2(\tilde{x}^P(\rho) - \omega_1) + 2(\bar{x}^P(\rho) - \omega_1)}{\beta + (1 - \pi_C)\sigma^2} \\
& \Longleftrightarrow \frac{2(\bar{x}^P(\rho) - \tilde{x}^P(\rho))}{\beta + (1 - \pi_C)\sigma^2} > 0.
\end{aligned}$$

So, ω_1 is minimized at $\omega_1 = 0$, and D1 requires placing a probability of 0 on any deviation from the $\omega_1 \in (0, \bar{x}^P(\rho))$ type as

$$D_\sigma(\omega_1 \in (0, \bar{x}^P(\rho)), \tilde{x}^P(\rho)) \subseteq D_\sigma(\omega_1 = 0, \tilde{x}^P(\rho)).$$

Next consider the incentive for the $\omega_1 \in (\underline{x}^P(\rho), 0)$ type to deviate to some $\tilde{x}^P(\rho) \in (0, \bar{x}^P(\rho))$. Then,

$$\mu(\tilde{x}^P(\rho)) > \frac{(\tilde{x}^P(\rho) - \omega_1) - (\underline{x}^P(\rho) - \omega_1) + \beta + (1 - \pi_C)\sigma^2}{\beta + (1 - \pi_C)\sigma^2}.$$

Taking the partial derivative WRT ω_1 ,

$$\begin{aligned}
\frac{\partial(\text{RHS})}{\partial\omega_1} &= \frac{-2(\tilde{x}^P(\rho) - \omega_1) + 2(\underline{x}^P(\rho) - \omega_1)}{\beta + (1 - \pi_C)\sigma^2} \\
&= \frac{2(\underline{x}^P(\rho) - \tilde{x}^P(\rho))}{\beta + (1 - \pi_C)\sigma^2} < 0.
\end{aligned}$$

as $\underline{x}^P(\rho) < 0$. Therefore, as ω_1 increases in $(\underline{x}^P(\rho), 0)$, the incentive to follow equilibrium behavior decreases. Because $\omega_1 \rightarrow 0$ converges to $\omega_1 = 0$ payoff, D1 requires placing a probability 0 on any $\tilde{x}^P(\rho) \in (0, \bar{x}^P(\rho))$ deviation coming from a $\omega_1 \in (\underline{x}^P(\rho), 0)$ type. An analogous argument shows the case for $\tilde{x}^P(\rho) \in (\underline{x}^P(\rho), 0)$. Therefore, D1 requires placing a probability 0 on any $\omega_1 \in (\underline{x}^P(\rho), \bar{x}^P(\rho)) \setminus \{0\}$ type.

Examining the $\omega_1 \in [0, \bar{x}^P(\rho)]$ case for the Uninformed incumbent. Now for the uninformed type's incentive to choose some $\tilde{x}^P(\rho) \in (0, \bar{x}^P(\rho))$. Their equilibrium payoff is

$$-\bar{x}^P(\rho)^2 - (2 - \rho)\sigma^2 + 2\beta.$$

If they deviate to $x_1 = \tilde{x}^P(\rho)$,

$$-\tilde{x}^P(\rho)^2 - \sigma^2 + \beta + \mu(\tilde{x}^P(\rho))(\beta - \sigma^2) - (1 - \mu(\tilde{x}^P(\rho))[(1 - \pi_C)\sigma^2].$$

The incentive compatibility constraint is

$$\begin{aligned} & -\tilde{x}^P(\rho)^2 - \sigma^2 + \beta + \mu(\tilde{x}^P(\rho))(\beta - \sigma^2) - (1 - \mu(\tilde{x}^P(\rho)))[(1 - \pi_C)\sigma^2] \\ & > -\bar{x}^P(\rho)^2 - (2 - \rho)\sigma^2 + 2\beta \\ & = \mu(\tilde{x}^P(\rho))(\beta - \sigma^2) - (1 - \mu(\tilde{x}^P(\rho)))[(1 - \pi_C)\sigma^2] \\ & > -\tilde{x}^P(\rho)^2 - \bar{x}^P(\rho)^2 + \sigma^2 - (2 - \rho)\sigma^2 + \beta \\ & = \frac{\mu(\tilde{x}^P(\rho))(\beta - \sigma^2 + (1 - \pi_C)\sigma^2)}{\beta - \sigma^2 + (1 - \pi_C)\sigma^2} \\ & > \frac{\tilde{x}^P(\rho)^2 - \bar{x}^P(\rho)^2 - (2 - \rho)\sigma^2 + \beta + (1 - \pi_C)\sigma^2}{\beta - \sigma^2 + (1 - \pi_C)\sigma^2} \\ & \iff \mu(\tilde{x}^P(\rho)) > \frac{\tilde{x}^P(\rho)^2 - \bar{x}^P(\rho)^2 - (2 - \rho)\sigma^2 + \beta + (1 - \pi_C)\sigma^2}{\beta - \sigma^2 + (1 - \pi_C)\sigma^2} \\ & = \mu(\tilde{x}^P(\rho)) > \frac{\tilde{x}^P(\rho)^2 - \bar{x}^P(\rho)^2 - \sigma^2 + \rho\sigma^2 + \beta + (1 - \pi_C)\sigma^2}{\beta - \sigma^2 + (1 - \pi_C)\sigma^2} \\ & = \mu(\tilde{x}^P(\rho)) > \frac{\tilde{x}^P(\rho)^2 - \bar{x}^P(\rho)^2 + \rho\sigma^2 + \beta - \pi_C\sigma^2}{\beta - \pi_C\sigma^2}. \end{aligned}$$

To show that the uninformed type is willing to deviate for a lower $\mu(\tilde{x}^P(\rho))$ than the $\omega_1 = 0$ type, set the incentive compatibility to

$$\frac{(\tilde{x}^P(\rho) - \omega_1) - (\bar{x}^P(\rho) - \omega_1) + \beta + (1 - \pi_C)\sigma^2}{\beta + (1 - \pi_C)\sigma^2} > \frac{\tilde{x}^P(\rho)^2 - \bar{x}^P(\rho)^2 + \rho\sigma^2 + \beta - \pi_C\sigma^2}{\beta - \pi_C\sigma^2}.$$

Setting $\omega_1 = 0$ reduces the incentive constraint to

$$\frac{\tilde{x}^P(\rho) - \bar{x}^P(\rho) + \beta + (1 - \pi_C)\sigma^2}{\beta + (1 - \pi_C)\sigma^2} > \frac{\tilde{x}^P(\rho)^2 - \bar{x}^P(\rho)^2 + \rho\sigma^2 + \beta - \pi_C\sigma^2}{\beta - \pi_C\sigma^2}.$$

Let $y = \tilde{x}^P(\rho) - \bar{x}^P(\rho)$. Then,

$$(\beta - \pi_C\sigma^2)(y + \beta + (1 - \pi_C)\sigma^2) > (\beta + (1 - \pi_C)\sigma^2)(y + \beta + \rho\sigma^2 - \pi_C\sigma^2).$$

This simplifies to

$$(\beta - \pi_C \sigma^2)(\beta + (1 - \pi_C)\sigma^2) > \sigma^2 y + \rho \sigma^2 + (\beta + (1 - \pi_C)\sigma^2)(\beta - \pi_C \sigma^2).$$

Further simplification results in

$$-\sigma^2 y - \rho \sigma^2 > (\beta + (1 - \pi_C)\sigma^2)(2\beta - \pi_C \sigma^2) - (\beta - \pi_C \sigma^2)(\beta + (1 - \pi_C)\sigma^2).$$

Therefore,

$$\sigma^2(\bar{x}^P(\rho) - \tilde{x}^P(\rho)) - \rho \sigma^2 > 0.$$

Because $\sigma^2 > 0$ and $\bar{x}^P(\rho) - \tilde{x}^P(\rho) > 0$, the incentive constraint holds for ρ sufficiently small. An analogous argument is clear for $\tilde{x}^P(\rho) \in (\underline{x}^P(\rho), 0)$. Therefore, if the voter puts probability 0 on an off-path policy choice $\tilde{x}^P(\rho) \in (\underline{x}^P(\rho), \bar{x}^P(\rho))$ coming from an uninformed type, these equilibria survive D1.

Step 2. Strategies and Beliefs form a Perfect Bayesian Equilibrium.

To show these strategies and beliefs form a Perfect Bayesian Equilibrium, I begin with the uninformed type.

Assume $d_1 = P$. Then, their utility is

$$-x^2 - \sigma^2$$

and choosing some $x_1 \in (\underline{x}^P(\rho), \bar{x}^P(\rho)) \setminus \{0\}$ results in a strictly worse payoff than selection $x_1 = 0$. Moreover, choosing some $x_1 > \bar{x}^P(\rho)$ or $x_1 < \underline{x}^P(\rho)$ and getting re-elected is strictly worse than setting $x_1 = \bar{x}^P(\rho)$ or $x_1 = \underline{x}^P(\rho)$. Both policy bounds make the uninformed type indifferent between $x_1 = \bar{x}^P(\rho)$, $x_1 = \underline{x}^P(\rho)$, and $x_1 = 0$. As such, the uninformed type mixes over the three actions.

For the informed type. If $\omega_1 \geq \bar{x}^P(\rho)$ or $\omega_1 \leq \underline{x}^P(\rho)$, the optimal policy choice is to select $x_1 = \omega_1$. Now, restrict $\omega_1 \in [0, \bar{x}^P(\rho))$. The optimal policy for re-election is $x_1 = \bar{x}^P(\rho)$. If the informed incumbent selects their bliss point, then $x_1 = \omega_1$ and they are removed from office. Their payoffs are

$$\beta - (1 - \pi_C)\sigma^2$$

for removal, and

$$-(\bar{x}^P(\rho) - \omega_1)^2 + 2\beta$$

for re-election. The cost of selecting $\bar{x}^P(\rho)$ is decreasing in ω_1 such that

$$\frac{\partial(\bar{x}^P(\rho) - \omega_1)^2 + 2\beta}{\partial\omega_1} = -2(\bar{x}^P(\rho) - \omega_1) < 0.$$

Therefore, if $\omega_1 = 0$ type prefers $\bar{x}^P(\rho)$ then so must all $\omega_1 \in (0, \bar{x}^P(\rho))$ types. Then,

$$-\bar{x}^P(\rho)^2 + 2\beta > \beta - (1 - \pi_C)\sigma^2 \iff \sigma^2 > \rho\sigma^2.$$

where $\bar{x}^P(\rho)^2 = \beta - \pi_C\sigma^2 + \rho\sigma^2$. This holds for $\rho \in [0, 1)$. The $\omega_1 \in (\underline{x}^P(\rho), 0)$ case is analogous.

As a result, the $\omega_1 = 0$ type prefers to select $x_1 = \bar{x}^P(\rho)$ and get re-elected over $x_1 = 0$. Because the $\omega_1 = 0$ type is willing, comparative statics show that all other $\omega_1 \in (0, \bar{x}^P(\rho))$ types are also willing. The loss from policy distortion is not strong enough to encourage selecting $x_1 = \omega_1$ for $\omega_1 \in [0, \bar{x}^P(\rho))$.

Finally, a voter must be willing to re-elect an incumbent following $x_1 \geq \bar{x}^P(\rho)$ or $x_1 \leq \underline{x}^P(\rho)$ and elect the challenger following $x_1 \in (\underline{x}^P(\rho), \bar{x}^P(\rho))$.

From D1, a policy $x_1 \in (\underline{x}^P(\rho), \bar{x}^P(\rho) \setminus \{0\})$ is off-the-path. Therefore, $\tilde{\pi}_I(x_1) \leq \pi_C$, and the voter elects the challenger. By D1, let $\tilde{\pi}_I(x_1) = 0$ for $x_1 \in (\underline{x}^P(\rho), \bar{x}^P(\rho) \setminus \{0\})$. The uninformed type selects $x_1 = 0$, so voters' posterior becomes $\tilde{\pi}_I(x_1) = 0 < \pi_C$. $x_1 > \bar{x}^P(\rho)$ and $x_1 < \underline{x}^P(\rho)$ are only selected by the informed incumbent. Under this assumption $\tilde{\pi}_I(x_1) = 1 > \pi_C$.

Let $x_1 = \bar{x}^P(\rho)$. In this case, $\tilde{\pi}_I(x_1) \geq \pi_C$. From Bayes Rule,

$$\tilde{\pi}_I(x_1) = \frac{\Pr(x_1 \mid \theta_I = H, d_1 = P) \cdot \pi_I}{\Pr(x_1 \mid \theta_I = H, d_1 = P) \cdot \pi_I + (1 - \pi_I) \Pr(x_1 \mid \theta_I = L, d_1 = P)}.$$

Let $\Pr(x_1 \mid \theta_I = H, d_1 = P) = F(\omega^P(\rho)) - F(0)$ and $\Pr(x_1 \mid \theta_I = L, d_1 = P) = \bar{\Sigma}$. Then,

$$\begin{aligned} \tilde{\pi}_I(x_1) &= \frac{[F(\omega^P(\rho)) - F(0)] \cdot \pi_I}{[F(\omega^P(\rho)) - F(0)] \cdot \pi_I + (1 - \pi_I) \bar{\Sigma}} \geq \pi_C \\ &\iff \frac{\pi_I(1 - \pi_C)}{\pi_C(1 - \pi_I)} (F(\omega^P(\rho)) - F(0)) \geq \bar{\Sigma}. \end{aligned}$$

A similar argument is made for $x_1 = \underline{x}^P(\rho)$. □

A.4 Voter Welfare

Remark 4. *Voter welfare satisfies:*

1. *Increasing the prior belief that the challenger is racially competent always increases welfare:*

$$\frac{\partial W}{\partial \pi_C} > 0.$$

2. *Increasing the prior belief that the incumbent is racially competent decreases welfare whenever*

$$\beta > (2 - \rho)\sigma^2.$$

Proof of Remark 4. Let voter welfare be defined as

$$\begin{aligned} W := & \pi_I \left\{ \rho \left[\int_{-\omega^P(\rho)}^0 -(\underline{x}^P(\rho) - \omega)^2 f(\omega) d\omega + \int_0^{\omega^P(\rho)} -(\bar{x}^P(\rho) - \omega)^2 f(\omega) d\omega \right] \right. \\ & \left. + (1 - \rho) \left[\int_{-\omega^P(\rho)}^0 -(\underline{x}^P(\rho) - \omega)^2 f(\omega) d\omega + \int_0^{\omega^P(\rho)} -(\bar{x}^P(\rho) - \omega)^2 f(\omega) d\omega \right] \right\} \\ & + (1 - \pi_I) \left\{ \rho \left[\int_{-\omega^P(\rho)}^0 -(\underline{x}^P(\rho) - \omega)^2 f(\omega) d\omega + \int_0^{\omega^P(\rho)} -(\bar{x}^P(\rho) - \omega)^2 f(\omega) d\omega \right] + (1 - \rho) [-\sigma^2 - (1 - \pi_C)\sigma^2] \right\}. \end{aligned}$$

1. $\frac{\partial W}{\partial \pi_C} > 0$. The derivative with respect to π_C is

$$(1 - \pi_C)(1 - \rho)\sigma^2 > 0.$$

As π_C increases, both $\omega^P(\rho)$ and $\bar{x}^P(\rho)$ decrease. This decreases the extent to which the incumbent will overreact. Moreover, it decreases $\bar{\Sigma}$ and $\underline{\Sigma}$, thereby reducing the incumbents' willingness to posture. Therefore, increasing π_C increases voter welfare.

2. $\frac{\partial W}{\partial \pi_I} < 0$. The derivative with respect to π_I is

$$(1 - \rho) \underbrace{\left[\int_{-\omega^P(\rho)}^0 -(\underline{x}^P(\rho) - \omega)^2 f(\omega) d\omega + \int_0^{\omega^P(\rho)} -(\bar{x}^P(\rho) - \omega)^2 f(\omega) d\omega + (2 - \pi_C)\sigma^2 \right]}_{=: \mathcal{L}(\rho) < 0},$$

where $\mathcal{L}(\rho)$ is the voter's expected policy loss within the semi-separating region—the cost borne when the policy $\bar{x}^P(\rho)$ or $\underline{x}^P(\rho)$ deviates from the true state ω . This term is strictly negative since it is a sum of squared losses weighted by the density. The expression is negative when $|\mathcal{L}(\rho)| > (2 - \pi_C)\sigma^2$, i.e., when the expected policy loss exceeds the welfare gain from avoiding the screening penalty. Since $|\mathcal{L}(\rho)|$ is increasing in $(\bar{x}^P(\rho))^2$, a sufficient condition is that the policy distortion itself exceeds the offset:

$$(\bar{x}^P(\rho))^2 > (2 - \pi_C)\sigma^2.$$

Substituting $(\bar{x}^P(\rho))^2 = \beta - \pi_C\sigma^2 + \rho\sigma^2$ yields

$$\beta - \pi_C\sigma^2 + \rho\sigma^2 > (2 - \pi_C)\sigma^2 \implies \beta > (2 - \rho)\sigma^2. \quad \square$$

B Comparative Statics

B.1 Impact of Racial Competency on Policy Extremity

Proposition 2 (Impact of ρ on Policy Extremity). *An increase in the probability that the second-period policy domain is universal leads to an increase in the extremity of the policy choice; that is,*

$$\frac{\partial \bar{x}^P(\rho)}{\partial \rho} > 0 \quad \text{and} \quad \frac{\partial \underline{x}^P(\rho)}{\partial \rho} < 0.$$

Furthermore, as $\rho \rightarrow 1$,

$$\lim_{\rho \rightarrow 1} \bar{x}^P(\rho) = \sqrt{\beta - (\pi_C - 1)\sigma^2} \quad \text{and} \quad \lim_{\rho \rightarrow 1} \underline{x}^P(\rho) = -\sqrt{\beta - (\pi_C - 1)\sigma^2}.$$

Proof of Proposition 2. Assume the equilibrium policy positions are given by

$$\bar{x}^P(\rho) = \sqrt{\beta - \pi_C \sigma^2 + \rho \sigma^2} \quad \text{and} \quad \underline{x}^P(\rho) = -\sqrt{\beta - \pi_C \sigma^2 + \rho \sigma^2}.$$

Define

$$g(\rho) = \beta - \pi_C \sigma^2 + \rho \sigma^2.$$

Then we can write

$$\bar{x}^P(\rho) = (g(\rho))^{1/2}.$$

Applying the chain rule,

$$\frac{d\bar{x}^P(\rho)}{d\rho} = \frac{d}{d\rho} (g(\rho)^{1/2}) = \frac{1}{2} g(\rho)^{-1/2} \cdot \frac{dg(\rho)}{d\rho}.$$

Since

$$\frac{dg(\rho)}{d\rho} = \sigma^2,$$

we have

$$\frac{\partial \bar{x}^P(\rho)}{\partial \rho} = \frac{1}{2} (\beta - \pi_C \sigma^2 + \rho \sigma^2)^{-1/2} \sigma^2 > 0.$$

Similarly, for the lower policy position,

$$\underline{x}^P(\rho) = -\sqrt{\beta - \pi_C \sigma^2 + \rho \sigma^2},$$

so that

$$\frac{\partial \underline{x}^P(\rho)}{\partial \rho} = -\frac{1}{2} (\beta - \pi_C \sigma^2 + \rho \sigma^2)^{-1/2} \sigma^2 < 0.$$

Taking the limit as $\rho \rightarrow 1$, we obtain

$$\lim_{\rho \rightarrow 1} \bar{x}^P(\rho) = \sqrt{\beta - \pi_C \sigma^2 + \sigma^2} = \sqrt{\beta - (\pi_C - 1)\sigma^2},$$

and

$$\lim_{\rho \rightarrow 1} \underline{x}^P(\rho) = -\sqrt{\beta - \pi_C \sigma^2 + \sigma^2} = -\sqrt{\beta - (\pi_C - 1)\sigma^2}.$$

□

B.2 Impact of Racial Competency on Winning Probability

Proposition 3 (Impact of ρ on the Winning Probability for the Uninformed Incumbent). *As ρ increases, the winning probability for the uninformed incumbent increases.*

Proof of Proposition 3. Assume that voters re-elect the incumbent if and only if the state ω falls in the interval $[-\omega^P(\rho), \omega^P(\rho)]$, where

$$\omega^P(\rho) = \sqrt{\beta - \pi_C \sigma^2 + \rho \sigma^2}.$$

Assume $\beta - \pi_C \sigma^2 + \rho \sigma^2 > 0$ so that $\omega^P(\rho) > 0$. Then the uninformed type's winning probability is

$$P_{win}^L(\rho) = \int_{-\omega^P(\rho)}^{\omega^P(\rho)} f(\omega) d\omega = F(\omega^P(\rho)) - F(-\omega^P(\rho)).$$

To show that $P_{win}^L(\rho)$ is increasing in ρ , we differentiate it with respect to ρ using partial derivatives. By the chain rule,

$$\frac{\partial P_{win}^L(\rho)}{\partial \rho} = \frac{\partial}{\partial \rho} [F(\omega^P(\rho)) - F(-\omega^P(\rho))] = F'(\omega^P(\rho)) \frac{\partial \omega^P(\rho)}{\partial \rho} - F'(-\omega^P(\rho)) \frac{\partial (-\omega^P(\rho))}{\partial \rho}.$$

Since $F'(x) = f(x)$ and $\frac{\partial (-\omega^P(\rho))}{\partial \rho} = -\frac{\partial \omega^P(\rho)}{\partial \rho}$, this becomes

$$\frac{\partial P_{win}^L(\rho)}{\partial \rho} = f(\omega^P(\rho)) \frac{\partial \omega^P(\rho)}{\partial \rho} + f(-\omega^P(\rho)) \frac{\partial \omega^P(\rho)}{\partial \rho} = [f(\omega^P(\rho)) + f(-\omega^P(\rho))] \frac{\partial \omega^P(\rho)}{\partial \rho}.$$

Next, differentiate $\omega^P(\rho)$ with respect to ρ . We have

$$\omega^P(\rho) = \left(\beta - \pi_C \sigma^2 + \rho \sigma^2 \right)^{1/2},$$

so that

$$\frac{\partial \omega^P(\rho)}{\partial \rho} = \frac{1}{2} \left(\beta - \pi_C \sigma^2 + \rho \sigma^2 \right)^{-1/2} \cdot \sigma^2 = \frac{\sigma^2}{2\omega^P(\rho)}.$$

Substituting back, we obtain

$$\frac{\partial P_{win}^L(\rho)}{\partial \rho} = [f(\omega^P(\rho)) + f(-\omega^P(\rho))] \frac{\sigma^2}{2\omega^P(\rho)}.$$

Since $f(\cdot) \geq 0$ it follows that,

$$\frac{\partial P_{win}^L(\rho)}{\partial \rho} > 0.$$

Thus, $P_{win}^L(\rho)$ is strictly increasing in ρ , which proves that as the probability of a universal second-period domain increases, so does the winning probability for the Uninformed incumbent. $\square$

B.3 Impact of Prior Beliefs on Mixing Probabilities

Proposition 4 (Impact of Prior Beliefs on Mixing Probabilities). *Let $\bar{\Sigma}$ and $\underline{\Sigma}$ denote the equilibrium mixing probabilities associated with right-posturing and left-posturing, respectively. Then,*

1. *An increase in the prior belief that the incumbent is racially competent raises the right-posturing mixing probability, $\frac{d\bar{\Sigma}}{d\pi_I} > 0$, and lowers the left-posturing mixing probability, $\frac{d\underline{\Sigma}}{d\pi_I} < 0$.*
2. *An increase in the prior belief that the challenger is racially competent lowers the right-posturing mixing probability, $\frac{d\bar{\Sigma}}{d\pi_C} < 0$, and raises the left-posturing mixing probability, $\frac{d\underline{\Sigma}}{d\pi_C} > 0$.*

Proof of Proposition 4. Let $x^P(\rho)$ be the equilibrium policy choice by the incumbent. We want to show that the posterior belief

$$\tilde{\pi}_I(x_1^P(\rho)) = \frac{\pi_I [F(\omega^P(\rho)) - F(0)]}{\pi_I [F(\omega^P(\rho)) - F(0)] + (1 - \pi_I) \Sigma}$$

is strictly increasing in π_I . Define $\omega^P(\rho) = x_1^P(\rho)$. Then

$$\frac{d}{d\pi_I} [\tilde{\pi}_I(x_1^P(\rho))] = \underbrace{\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \pi_I} (\omega^P(\rho))}_{\text{direct}} + \underbrace{\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial x_1^P(\rho)} \frac{dx_1^P(\rho)}{d\pi_I}}_{\text{indirect}}.$$

1. Direct effect. Holding $\omega^P(\rho)$ fixed, write

$$\begin{aligned} \frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \pi_I} = & \frac{\left(\frac{\partial}{\partial \pi_I} [\pi_I (F(\omega^P(\rho)) - F(0))] \right) [\pi_I (F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \Sigma] - [\pi_I (F(\omega^P(\rho)) - F(0))] \left(\frac{\partial}{\partial \pi_I} [\pi_I (F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \Sigma] \right)}{[\pi_I (F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \Sigma]^2}. \end{aligned}$$

Now compute each partial derivative:

$$\frac{\partial}{\partial \pi_I} [\pi_I (F(\omega^P(\rho)) - F(0))] = F(\omega^P(\rho)) - F(0),$$

$$\frac{\partial}{\partial \pi_I} [\pi_I (F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \Sigma] = (F(\omega^P(\rho)) - F(0)) - \Sigma.$$

Therefore,

$$\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \pi_I} = \frac{[F(\omega^P(\rho)) - F(0)] [\pi_I (F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \Sigma] - [\pi_I (F(\omega^P(\rho)) - F(0))] [(F(\omega^P(\rho)) - F(0)) - \Sigma]}{[\pi_I (F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \Sigma]^2}.$$

Break the numerator into two parts:

$$\begin{aligned} & [F(\omega^P(\rho)) - F(0)] [\pi_I (F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \Sigma] \\ & - [\pi_I (F(\omega^P(\rho)) - F(0))] [(F(\omega^P(\rho)) - F(0)) - \Sigma] \\ = & \underbrace{\pi_I (F(\omega^P(\rho)) - F(0))^2 + (1 - \pi_I) (F(\omega^P(\rho)) - F(0)) \Sigma}_{\text{first term}} \\ & - \underbrace{\pi_I (F(\omega^P(\rho)) - F(0))^2 - \pi_I (F(\omega^P(\rho)) - F(0)) \Sigma}_{\text{second term}} \\ = & [\pi_I (F(\omega^P(\rho)) - F(0))^2 + (1 - \pi_I) (F(\omega^P(\rho)) - F(0)) \Sigma] \\ & - [\pi_I (F(\omega^P(\rho)) - F(0))^2 - \pi_I (F(\omega^P(\rho)) - F(0)) \Sigma] \\ = & [F(\omega^P(\rho)) - F(0)] \Sigma. \end{aligned}$$

Hence

$$\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \pi_I} = \frac{[F(\omega^P(\rho)) - F(0)] \Sigma}{[\pi_I (F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \Sigma]^2}.$$

2. Indirect effect. First compute $\partial \tilde{\pi}_I(x_1^P(\rho))/\partial x_1^P(\rho)$ with $\omega^P(\rho) = x_1^P(\rho)$ held fixed in the denominator:

$$\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial x_1^P(\rho)} = \frac{[\pi_I (F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \Sigma] \pi_I f(\omega^P(\rho)) - \pi_I (F(\omega^P(\rho)) - F(0)) \pi_I f(\omega^P(\rho))}{[\pi_I (F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \Sigma]^2}.$$

Factor $\pi_I f(\omega^P(\rho))$ in the numerator:

$$\begin{aligned} & \pi_I f(\omega^P(\rho)) \left[(\pi_I (F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \Sigma) - \pi_I (F(\omega^P(\rho)) - F(0)) \right] \\ = & \pi_I f(\omega^P(\rho)) (1 - \pi_I) \Sigma = \pi_I (1 - \pi_I) \Sigma f(\omega^P(\rho)). \end{aligned}$$

Thus

$$\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial x_1^P(\rho)} = \frac{\pi_I (1 - \pi_I) \Sigma f(\omega^P(\rho))}{[\pi_I (F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \Sigma]^2}.$$

Finally, since $x_1^P(\rho) = \omega^P(\rho)$ does *not* depend on π_I ,

$$\frac{dx_1^P(\rho)}{d\pi_I} = 0,$$

so the indirect effect is

$$\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial x_1^P(\rho)} \cdot \frac{dx_1^P(\rho)}{d\pi_I} = \frac{\pi_I(1 - \pi_I) \Sigma f(\omega^P(\rho))}{\left[\pi_I(F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \Sigma \right]^2} \cdot 0 = 0.$$

3. Total derivative. Summing direct and indirect effects,

$$\begin{aligned} \frac{d}{d\pi_I} [\tilde{\pi}_I(x_1^P(\rho))] &= \frac{(F(\omega^P(\rho)) - F(0)) \Sigma}{\underbrace{\left[\pi_I(F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \Sigma \right]^2}_{\text{direct}}} + 0 \\ &= \frac{(F(\omega^P(\rho)) - F(0)) \Sigma}{\left[\pi_I(F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \Sigma \right]^2} > 0. \end{aligned}$$

Therefore, $\tilde{\pi}_I(x_1^P(\rho))$ is strictly increasing in π_I .

4. To show $\frac{d\bar{\Sigma}}{d\pi_I} > 0$.

Now, suppose $\tilde{\pi}_I(x_1^P(\rho)) = \pi_C$. We established that $\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \pi_I} > 0$. Now, to calculate the impact of π_I on $\bar{\Sigma}$ we use the Implicit Function Theorem which states that

$$\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \pi_I} + \frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \bar{\Sigma}} \frac{d\bar{\Sigma}}{d\pi_I} = 0.$$

First, to determine the sign of $\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \bar{\Sigma}}$, we use the quotient rule:

$$\begin{aligned} \frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \bar{\Sigma}} &= \frac{\pi_I(F(\omega^P(\rho)) - F(0))}{\pi_I(F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \bar{\Sigma}} \\ &\iff -\frac{1 - \pi_I}{[\pi_I(F(\omega^P(\rho)) - F(0)) + (1 - \pi_I) \bar{\Sigma}]^2} < 0. \end{aligned}$$

Then, subtract $\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \pi_I}$ from the LHS:

$$\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \bar{\Sigma}} \frac{d\bar{\Sigma}}{d\pi_I} = -\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \pi_I}.$$

After, divide $\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \bar{\Sigma}}$ from the LHS:

$$\frac{d\bar{\Sigma}}{d\pi_I} = -\frac{\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \pi_I}}{\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \bar{\Sigma}}}.$$

Because $\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \bar{\Sigma}} < 0$ and $\frac{\partial \tilde{\pi}_I(x_1^P(\rho))}{\partial \pi_I} > 0$, $\frac{d\bar{\Sigma}}{d\pi_I} > 0$. Therefore, increasing the prior belief that an incumbent is high-type increases $\bar{\Sigma}$. An analogous case is made for $\underline{\Sigma}$.

5. To show $\frac{d\bar{\Sigma}}{d\pi_C} < 0$.

First, we solve the state threshold with respect to π_C . Let $\omega^P(\rho) = \sqrt{\beta - \pi_C \sigma^2 + \rho \sigma^2}$. We want to show that

$$\frac{\partial \omega^P(\rho)}{\partial \pi_C} < 0.$$

By the chain rule, let

$$\omega^P(\pi_C) = (f(\pi_C))^{\frac{1}{2}}.$$

Then,

$$\frac{d\omega^P(\rho)}{d\pi_C} = \frac{1}{2}(f(\pi_C))^{-\frac{1}{2}},$$

and

$$\frac{d(\beta - \pi_C \sigma^2 + \rho \sigma^2)}{d\pi_C} = -\sigma^2.$$

Multiplying results in

$$\frac{\partial \omega^P(\rho)}{\partial \pi_C} = -\frac{\sigma^2}{2\omega^P(\rho)} < 0.$$

Suppose $\tilde{\pi}_I(x^P(\rho)) = \pi_C$. Then, simple algebra results in

$$\bar{\Sigma} = \frac{\pi_I(1 - \pi_C)}{\pi_C(1 - \pi_I)}(F(\omega^P(\rho)) - F(0)).$$

We want to show that $\frac{\partial \bar{\Sigma}}{\partial \pi_C} < 0$. To do so, we will utilize the multivariate chain rule. First, the direct effect:

$$\begin{aligned} \frac{\partial \bar{\Sigma}}{\partial \pi_C} &= \frac{\pi_I}{1 - \pi_I} \cdot \frac{d}{d\pi_C} \left(\frac{1 - \pi_C}{\pi_C} \right) \\ \iff &\frac{\pi_I}{1 - \pi_I} \cdot -\frac{1}{(\pi_C)^2} \cdot F(\omega^P) - F(0) \\ &= -\frac{\pi_I(F(\omega^P(\rho)) - F(0))}{(\pi_C)^2}. \end{aligned}$$

Next, the indirect effect:

$$\frac{\partial \bar{\Sigma}}{\partial \omega^P(\rho)} = \frac{\pi_I(1 - \pi_C)}{\pi_C(1 - \pi_I)} f(\omega^P(\rho)),$$

and

$$\frac{\partial \omega^P(\rho)}{\partial \pi_C} = -\frac{\sigma^2}{2\omega^P},$$

from above. Then, to calculate the total derivative via the multivariate chain rule:

$$\frac{d\bar{\Sigma}}{d\pi_C} = -\frac{\pi_I(F(\omega^P(\rho)) - F(0))}{(\pi_C)^2} + \frac{\pi_I(1 - \pi_C)}{\pi_C(1 - \pi_I)} f(\omega^P(\rho)) \cdot -\frac{\sigma^2}{2\omega^P} < 0.$$

The result of the multivariate chain rule is that $\frac{d\bar{\Sigma}}{d\pi_C} < 0$. Therefore, increasing the prior belief that a challenger is high-type decreases $\bar{\Sigma}$. An analogous case is made for $\underline{\Sigma}$. $\square$

C Extensions

C.1 Racial Gerrymandering

Proposition 5 (Racial Gerrymandering). *Suppose the district faces redistricting uncertainty $\tau \in [0, 1]$. Then,*

1. **Policy Extremity.** *An increase in the probability that the district is cracked leads to an increase in the extremity of the policy choice:*

$$\frac{\partial \bar{x}^P(\tau)}{\partial \tau} > 0 \quad \text{and} \quad \frac{\partial \underline{x}^P(\tau)}{\partial \tau} < 0.$$

2. **Winning Probability.** *As τ increases, the winning probability for the uninformed incumbent increases:*

$$\frac{\partial P_{win}^L}{\partial \tau} > 0.$$

3. **Interaction effect on the state threshold.** *The marginal benefit of cracking on the state threshold increases as ρ increases:*

$$\frac{\partial^2 \omega^P(\rho, \tau)}{\partial \tau \partial \rho} > 0,$$

when β is sufficiently large.

4. **Interaction effect on the winning probability.** *The marginal benefit of cracking on the incumbent's winning probability increases as ρ increases:*

$$\frac{\partial^2 P_{win}^L}{\partial \tau \partial \rho} > 0,$$

when β is sufficiently large.

Proof of Proposition 5.

1. **To show** $\frac{\partial \bar{x}^P(\tau)}{\partial \tau} > 0$ **and** $\frac{\partial \underline{x}^P(\tau)}{\partial \tau} < 0$.

Assume the equilibrium policy positions are given by

$$\bar{x}^P(\tau) = \sqrt{\beta - \pi_C \sigma^2 + \rho \sigma^2 + \tau \sigma^2 + \rho \tau \sigma^2} \quad \text{and} \quad \underline{x}^P(\tau) = -\sqrt{\beta - \pi_C \sigma^2 + \rho \sigma^2 + \tau \sigma^2 + \rho \tau \sigma^2}.$$

Define

$$g(\tau) = \beta - \pi_C \sigma^2 + \rho \sigma^2 + \tau \sigma^2 + \rho \tau \sigma^2.$$

Then we can write

$$\bar{x}^P(\tau) = (g(\tau))^{1/2}.$$

Applying the chain rule,

$$\frac{d\bar{x}^P(\tau)}{d\tau} = \frac{d}{d\tau} (g(\tau)^{1/2}) = \frac{1}{2} g(\tau)^{-1/2} \cdot \frac{dg(\tau)}{d\tau}.$$

Since

$$\frac{dg(\tau)}{d\tau} = (1 + \rho) \sigma^2,$$

we have

$$\frac{\partial \bar{x}^P(\tau)}{\partial \tau} = \frac{1}{2} (\beta - \pi_C \sigma^2 + \rho \sigma^2 + \tau \sigma^2 + \rho \tau \sigma^2)^{-1/2} (1 + \rho) \sigma^2 > 0.$$

Similarly, for the lower policy position,

$$\underline{x}^P(\tau) = -\sqrt{\beta - \pi_C \sigma^2 + \rho \sigma^2 + \tau \sigma^2 + \rho \tau \sigma^2},$$

so that

$$\frac{\partial \underline{x}^P(\tau)}{\partial \tau} = -\frac{1}{2} (\beta - \pi_C \sigma^2 + \rho \sigma^2 + \tau \sigma^2 + \rho \tau \sigma^2)^{-1/2} (1 + \rho) \sigma^2 < 0.$$

2. **To show** $\frac{\partial P_{win}^L}{\partial \tau} > 0$.

Define the winning probability for the low-type as

$$P_{win}^L(\tau) = \int_{-\omega^P(\tau)}^{\omega^P(\tau)} f(\omega) d\omega = F(\omega^P(\tau)) - F(-\omega^P(\tau)).$$

We want to show that $\frac{\partial P_{win}^L(\tau)}{\partial \tau} > 0$.

To show that $P_{win}^L(\tau)$ is increasing in τ , we differentiate with respect to τ . By the chain rule,

$$\begin{aligned}\frac{\partial P_{win}^L(\tau)}{\partial \tau} &= \frac{\partial}{\partial \tau} [F(\omega^P(\tau)) - F(-\omega^P(\tau))] \\ &= \left[F'(\omega^P(\tau)) \frac{\partial \omega^P(\tau)}{\partial \tau} - F'(-\omega^P(\tau)) \frac{\partial (-\omega^P(\tau))}{\partial \tau} \right].\end{aligned}$$

Since $F'(x) = f(x)$ and $\frac{\partial (-\omega^P(\tau))}{\partial \tau} = -\frac{\partial \omega^P(\tau)}{\partial \tau}$, this becomes

$$\frac{\partial P_{win}^L(\tau)}{\partial \tau} = [f(\omega^P(\tau)) + f(-\omega^P(\tau))] \frac{\partial \omega^P(\tau)}{\partial \tau}.$$

From Part 1, we have

$$\omega^P(\tau) = \sqrt{\beta - \pi_C \sigma^2 + \rho \sigma^2 + \tau \sigma^2 + \rho \tau \sigma^2},$$

so that

$$\frac{\partial \omega^P(\tau)}{\partial \tau} = \frac{(1 + \rho)\sigma^2}{2\omega^P(\tau)}.$$

Substituting back, we obtain

$$\frac{\partial P_{win}^L(\tau)}{\partial \tau} = [f(\omega^P(\tau)) + f(-\omega^P(\tau))] \frac{(1 + \rho)\sigma^2}{2\omega^P(\tau)}.$$

By assumption, $\beta - \pi_C \sigma^2 + \rho \sigma^2 + \tau \sigma^2 + \rho \tau \sigma^2 > 0$. Moreover, $f(\cdot) \geq 0$ and $(1 + \rho) > 0$. Then, it follows that

$$\frac{\partial P_{win}^L(\tau)}{\partial \tau} > 0.$$

Thus, $P_{win}^L(\tau)$ is strictly increasing in τ , which proves that as the probability of a cracked district increases, so does the winning probability for the uninformed incumbent.

3. To show $\frac{\partial^2 \omega^P(\tau)}{\partial \tau \partial \rho} > 0$ when β is sufficiently large.

Given that

$$\frac{\partial \omega^P(\tau)}{\partial \tau} = \frac{(1 + \rho)\sigma^2}{2\omega^P(\tau)},$$

the cross-partial is

$$\frac{\partial^2 \omega^P(\tau)}{\partial \tau \partial \rho} = \frac{\partial \left[\frac{(1 + \rho)\sigma^2}{2\omega^P(\tau)} \right]}{\partial \rho}.$$

By the quotient rule, let $N = (1 + \rho)\sigma^2$ and $D = 2\omega^P(\tau)$. Then $N' = \sigma^2$ and $D' = 2 \cdot \frac{\partial \omega^P(\tau)}{\partial \rho}$, and

$$\frac{\partial^2 \omega^P(\tau)}{\partial \tau \partial \rho} = \frac{D \cdot N' - N \cdot D'}{D^2}$$

$$\begin{aligned}
&= \frac{2\omega^P(\tau) \cdot \sigma^2 - (1 + \rho)\sigma^2 \cdot 2\frac{\partial\omega^P(\tau)}{\partial\rho}}{(2\omega^P(\tau))^2} \\
&= \frac{2\sigma^2\omega^P(\tau)}{4(\omega^P(\tau))^2} - \frac{2(1 + \rho)\sigma^2}{4(\omega^P(\tau))^2} \cdot \frac{\partial\omega^P(\tau)}{\partial\rho} \\
&= \frac{\sigma^2}{2\omega^P(\tau)} - \frac{(1 + \rho)\sigma^2}{2(\omega^P(\tau))^2} \cdot \frac{(1 + \tau)\sigma^2}{2\omega^P(\tau)} \\
&= \frac{\sigma^2}{2\omega^P(\tau)} - \frac{(1 + \rho)(1 + \tau)\sigma^4}{4(\omega^P(\tau))^3}.
\end{aligned}$$

Combining over a common denominator of $4(\omega^P(\tau))^3$:

$$\begin{aligned}
&= \frac{2\sigma^2(\omega^P(\tau))^2 - (1 + \rho)(1 + \tau)\sigma^4}{4(\omega^P(\tau))^3} \\
&= \frac{\sigma^2 [2(\omega^P(\tau))^2 - (1 + \rho)(1 + \tau)\sigma^2]}{4(\omega^P(\tau))^3}.
\end{aligned}$$

Substituting $(\omega^P(\tau))^2 = \beta - \pi_C\sigma^2 + \rho\sigma^2 + \tau\sigma^2 + \rho\tau\sigma^2$ and expanding $(1 + \rho)(1 + \tau) = 1 + \rho + \tau + \rho\tau$:

$$\begin{aligned}
&= \frac{\sigma^2 [2\beta - 2\pi_C\sigma^2 + 2\rho\sigma^2 + 2\tau\sigma^2 + 2\rho\tau\sigma^2 - (1 + \rho + \tau + \rho\tau)\sigma^2]}{4(\omega^P(\tau))^3} \\
&= \frac{\sigma^2 [2\beta + (\rho + \tau + \rho\tau - 2\pi_C - 1)\sigma^2]}{4(\omega^P(\tau))^3}.
\end{aligned}$$

The denominator is strictly positive. Thus, the sign of the cross-partial depends on the numerator bracket

$$2\beta + (\rho + \tau + \rho\tau - 2\pi_C - 1)\sigma^2.$$

When β is sufficiently large—specifically, when

$$\beta > \frac{(1 + 2\pi_C - \rho - \tau - \rho\tau)\sigma^2}{2},$$

the bracket is positive, and therefore

$$\frac{\partial^2\omega^P(\tau)}{\partial\tau\partial\rho} > 0,$$

implying that the marginal effect of cracking on the state threshold increases as ρ increases.

4. To show $\frac{\partial^2 P_{win}^L}{\partial\tau\partial\rho} > 0$ when β is sufficiently large.

Assume f is symmetric about zero and single-peaked at zero, so that $f'(\omega) \leq 0$ for $\omega \geq 0$ and $f'(-\omega) = -f'(\omega)$. Define the winning probability for the low-type as

$$P_{win}^L(\tau) = \int_{-\omega^P(\tau)}^{\omega^P(\tau)} f(\omega) d\omega = F(\omega^P(\tau)) - F(-\omega^P(\tau)).$$

We want to show that $\frac{\partial^2 P_{win}^L(\tau)}{\partial \tau \partial \rho} > 0$. First, we know that

$$\frac{\partial P_{win}^L(\tau)}{\partial \tau} = [f(\omega^P(\tau)) + f(-\omega^P(\tau))] \frac{\partial \omega^P(\tau)}{\partial \tau}.$$

Now, to locate $\frac{\partial^2 P_{win}^L(\tau)}{\partial \tau \partial \rho}$,

$$\frac{\partial^2 P_{win}^L(\tau)}{\partial \tau \partial \rho} = \frac{\partial [[f(\omega^P(\tau)) + f(-\omega^P(\tau))] \frac{\partial \omega^P(\tau)}{\partial \tau}]}{\partial \rho}.$$

Let $A := [f(\omega^P(\tau)) + f(-\omega^P(\tau))]$ and $B := \frac{\partial \omega^P(\tau)}{\partial \tau}$. First,

$$\begin{aligned} \frac{\partial A}{\partial \rho} &= \left[f'(\omega^P(\tau)) \frac{\partial \omega^P(\tau)}{\partial \rho} + f'(-\omega^P(\tau)) \frac{\partial (-\omega^P(\tau))}{\partial \rho} \right] \\ &= [f'(\omega^P(\tau)) - f'(-\omega^P(\tau))] \frac{\partial \omega^P(\tau)}{\partial \rho}. \end{aligned}$$

Second,

$$\frac{\partial B}{\partial \rho} = \frac{\partial [\frac{\partial \omega^P(\tau)}{\partial \tau}]}{\partial \rho} = \frac{\partial^2 \omega^P(\tau)}{\partial \tau \partial \rho},$$

which, from Part 3, is equal to $\frac{\sigma^2 [2\beta + (\rho + \tau + \rho\tau - 2\pi_C - 1)\sigma^2]}{4(\omega^P(\tau))^3}$. By the product rule,

$$\begin{aligned} \frac{\partial^2 P_{win}^L(\tau)}{\partial \tau \partial \rho} &= \frac{\partial A}{\partial \rho} \cdot B + A \frac{\partial B}{\partial \rho} \\ &= [f'(\omega^P(\tau)) - f'(-\omega^P(\tau))] \frac{\partial \omega^P(\tau)}{\partial \rho} \frac{\partial \omega^P(\tau)}{\partial \tau} + [f(\omega^P(\tau)) + f(-\omega^P(\tau))] \frac{\partial^2 \omega^P(\tau)}{\partial \tau \partial \rho}. \end{aligned}$$

Rearrange and simplify,

$$[f(\omega^P(\tau)) + f(-\omega^P(\tau))] \frac{\partial^2 \omega^P(\tau)}{\partial \tau \partial \rho} + [f'(\omega^P(\tau)) - f'(-\omega^P(\tau))] \frac{\partial \omega^P(\tau)}{\partial \rho} \frac{\partial \omega^P(\tau)}{\partial \tau}.$$

We know that

$$\frac{\partial \omega^P(\tau)}{\partial \tau} > 0 \quad \frac{\partial \omega^P(\tau)}{\partial \rho} > 0,$$

and, when β is sufficiently large (specifically, $\beta > \frac{(1+2\pi_C - \rho - \tau - \rho\tau)\sigma^2}{2}$),

$$\frac{\partial^2 \omega^P(\tau)}{\partial \tau \partial \rho} > 0,$$

so the first term is positive. For the second term, symmetry and single-peakedness give $f'(\omega^P(\tau)) - f'(-\omega^P(\tau)) = 2f'(\omega^P(\tau)) \leq 0$, so this term is non-positive but bounded, while the first term grows linearly in β . Hence, for β sufficiently large, the first term dominates and

$$\frac{\partial^2 P_{win}^L(\tau)}{\partial \tau \partial \rho} > 0,$$

implying that the marginal effect of cracking on the winning probability increases as ρ increases. $\square$

C.1.1 Voter Welfare

Proposition 6 (Voter Welfare under Racial Gerrymandering). *Voter welfare satisfies:*

1. *Welfare is decreasing in the probability of cracking:*

$$\frac{\partial W}{\partial \tau} < 0.$$

2. *The marginal effect of cracking on welfare is decreasing in ρ :*

$$\frac{\partial^2 W}{\partial \tau \partial \rho} < 0.$$

Proof of Proposition 6. Define the expected policy loss within the semi-separating region:

$$\mathcal{L}(\rho, \tau) := \int_{-\omega^P(\rho, \tau)}^0 -(\underline{x}^P(\rho, \tau) - \omega)^2 f(\omega) d\omega + \int_0^{\omega^P(\rho, \tau)} -(\bar{x}^P(\rho, \tau) - \omega)^2 f(\omega) d\omega.$$

Using this notation, voter welfare can be written compactly as

$$W(\rho, \tau) = \pi_I \mathcal{L}(\rho, \tau) + (1 - \pi_I) \left\{ \rho \mathcal{L}(\rho, \tau) + (1 - \rho) [-\sigma^2 - (1 - \pi_C) \sigma^2] \right\}. \quad (1)$$

Part 1. Differentiating (1) with respect to τ :

$$\frac{\partial W}{\partial \tau} = \pi_I \frac{\partial \mathcal{L}}{\partial \tau} + (1 - \pi_I) \rho \frac{\partial \mathcal{L}}{\partial \tau}. \quad (2)$$

Since the payoff from selecting the challenger $-\sigma^2 - (1 - \pi_C) \sigma^2$ does not depend on τ , the only channel through which τ affects welfare is via the expected policy loss $\mathcal{L}(\rho, \tau)$. It remains to sign $\partial \mathcal{L} / \partial \tau$.

Define

$$h_-(\omega, \rho, \tau) := -(\underline{x}^P(\rho, \tau) - \omega)^2 f(\omega), \quad h_+(\omega, \rho, \tau) := -(\bar{x}^P(\rho, \tau) - \omega)^2 f(\omega).$$

Then, using Leibniz's integral rule,

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \tau} &= \underbrace{h_-(-\omega^P, \rho, \tau) \frac{\partial \omega^P}{\partial \tau}}_{\text{boundary: lower limit}} + \underbrace{h_+(\omega^P, \rho, \tau) \frac{\partial \omega^P}{\partial \tau}}_{\text{boundary: upper limit}} \\ &\quad + \int_{-\omega^P}^0 \frac{\partial h_-}{\partial \tau} d\omega + \int_0^{\omega^P} \frac{\partial h_+}{\partial \tau} d\omega, \end{aligned} \quad (3)$$

where $\frac{\partial}{\partial \tau}(-\omega^P) = -\frac{\partial \omega^P}{\partial \tau}$.

Signing the boundary terms. The equilibrium policy bounds satisfy $\bar{x}^P = \omega^P$ and $\underline{x}^P = -\omega^P$. At the boundaries $\omega = \pm\omega^P$, and:

$$(\bar{x}^P - \omega^P)^2 = (\omega^P - \omega^P)^2 = 0, \quad (\underline{x}^P - (-\omega^P))^2 = (-\omega^P + \omega^P)^2 = 0.$$

Hence $h_+(\omega^P, \rho, \tau) = h_-(-\omega^P, \rho, \tau) \equiv 0$, and the boundary terms in (3) vanish identically. The sign of $\partial\mathcal{L}/\partial\tau$ is therefore determined entirely by the interior integrals.

Signing the interior terms. The interior derivatives capture the effect of increased policy extremity on the loss within the existing semi-separating region:

$$\frac{\partial h_-}{\partial\tau} = -2(\underline{x}^P - \omega) \frac{\partial \underline{x}^P}{\partial\tau} f(\omega), \quad \frac{\partial h_+}{\partial\tau} = -2(\bar{x}^P - \omega) \frac{\partial \bar{x}^P}{\partial\tau} f(\omega).$$

From Proposition 5, $\frac{\partial \bar{x}^P}{\partial\tau} > 0$ and $\frac{\partial \underline{x}^P}{\partial\tau} < 0$. For $\omega \in [0, \omega^P]$, we have $\bar{x}^P > \omega \geq 0$, so $(\bar{x}^P - \omega) > 0$ and the integrand is negative. By symmetry, for $\omega \in [-\omega^P, 0]$, $(\underline{x}^P - \omega) < 0$ and $\frac{\partial \underline{x}^P}{\partial\tau} < 0$, so the integrand is again negative. Therefore, both interior integrals are strictly negative.

Combining: $\frac{\partial \mathcal{L}}{\partial\tau} < 0$. Substituting back into (2), both terms on the right-hand side are negative since $\pi_I > 0$ and $(1 - \pi_I)\rho \geq 0$. Hence $\frac{\partial W}{\partial\tau} < 0$. $\square$

Part 2. Differentiating (2) with respect to ρ :

$$\frac{\partial^2 W}{\partial\tau \partial\rho} = [\pi_I + (1 - \pi_I)\rho] \frac{\partial^2 \mathcal{L}}{\partial\tau \partial\rho} + (1 - \pi_I) \frac{\partial \mathcal{L}}{\partial\tau}. \quad (4)$$

The bracketed weight is strictly positive and $(1 - \pi_I) \frac{\partial \mathcal{L}}{\partial\tau} < 0$ from Part 1; it remains to sign $\frac{\partial^2 \mathcal{L}}{\partial\tau \partial\rho}$.

Boundary and threshold channels vanish. Differentiating (3) with respect to ρ , both the boundary and threshold cross-partial terms vanish by the same argument as Part 1, since $h_{\pm}(\pm\omega^P, \rho, \tau) \equiv 0$. The expression reduces to the interior terms:

$$\frac{\partial^2 \mathcal{L}}{\partial\tau \partial\rho} = \frac{\partial}{\partial\rho} \left[\int_{-\omega^P}^0 \frac{\partial h_-}{\partial\tau} d\omega + \int_0^{\omega^P} \frac{\partial h_+}{\partial\tau} d\omega \right].$$

Interior terms. Recall $\frac{\partial h_+}{\partial\tau} = -2(\bar{x}^P - \omega) \frac{\partial \bar{x}^P}{\partial\tau} f(\omega)$. Differentiating with respect to ρ via the product rule yields

$$\frac{\partial^2 h_+}{\partial\tau \partial\rho} = -2f(\omega) \left[\frac{\partial \bar{x}^P}{\partial\rho} \frac{\partial \bar{x}^P}{\partial\tau} + (\bar{x}^P - \omega) \frac{\partial^2 \bar{x}^P}{\partial\tau \partial\rho} \right].$$

Since $\bar{x}^P = \omega^P$, the policy-bound cross-partial inherits the sign of $\frac{\partial^2 \omega^P}{\partial\tau \partial\rho}$. By Proposition 5 Part 3, when $\beta > \frac{(1+2\pi_C - \rho - \tau - \rho\tau)\sigma^2}{2}$,

$$\frac{\partial^2 \omega^P}{\partial\tau \partial\rho} = \frac{\sigma^2 [2\beta + (\rho + \tau + \rho\tau - 2\pi_C - 1)\sigma^2]}{4(\omega^P)^3} > 0.$$

On the interior of the upper region, $\bar{x}^P - \omega = \omega^P - \omega > 0$, so the bracketed term is positive and $\frac{\partial^2 h_+}{\partial \tau \partial \rho} < 0$. By symmetry, the lower integral contributes equivalently. Hence $\frac{\partial^2 \mathcal{L}}{\partial \tau \partial \rho} < 0$ whenever $\beta > \frac{(1+2\pi_C - \rho - \tau - \rho\tau)\sigma^2}{2}$.

Combining. Substituting back into (4):

$$\frac{\partial^2 W}{\partial \tau \partial \rho} = \underbrace{[\pi_I + (1 - \pi_I)\rho]}_{<0} \frac{\partial^2 \mathcal{L}}{\partial \tau \partial \rho} + \underbrace{(1 - \pi_I)}_{<0} \frac{\partial \mathcal{L}}{\partial \tau}.$$

Therefore,

$$\frac{\partial^2 W}{\partial \tau \partial \rho} < 0 \quad \text{when} \quad \beta > \frac{(1 + 2\pi_C - \rho - \tau - \rho\tau)\sigma^2}{2}.$$

□

C.2 Legislative Constraints

C.2.1 When $d_1 = U$.

Lemma 3. (Symmetric Legislative Constraint). *Suppose the legislative constraints are strong ($\Psi < \bar{x}^P(\rho)$).*

1. *Assume the incumbent is popular ($\pi_I > \pi_C$).*

- *Informed Incumbent: If $\omega \geq 0$ then the incumbent chooses $x_1 = \Psi$. If $\omega < 0$, then the incumbent chooses $x_1 = -\Psi$.*
- *Uninformed Incumbent: The incumbent chooses $x_1 = \Psi$ with probability $1 - F(0)$ and $x = -\Psi$ with probability $F(0)$.*
- *Voter: The voter always re-elects the incumbent.*

2. *Assume the incumbent is unpopular ($\pi_I < \pi_C$).*

- *Informed Incumbent: If $\omega \geq 0$ then the incumbent chooses $x_1 = \Psi$. If $\omega < 0$, then the incumbent chooses $x_1 = -\Psi$.*
- *Uninformed Incumbent: The incumbent chooses $x_1 = \Psi$ with probability $\frac{\pi_I(1-\pi_C)}{\pi_C(1-\pi_I)}(1 - F(0))$ and $x = -\Psi$ with probability $\frac{\pi_I(1-\pi_C)}{\pi_C(1-\pi_I)}F(0)$.*
- *Voter: The voter reelects the incumbent with probability $\mu(\Psi) = \frac{\Psi^2 + \sigma^2}{\beta - (1-\rho)\sigma^2 - V_C(\Psi)}$.*

Proof of Lemma 3.

1. Suppose $\pi_I > \pi_C$. The updated belief that the incumbent is high-type is

$$\tilde{\pi}_I(\Psi) = \frac{\pi_I(1 - F(0))}{\pi_I(1 - F(0)) + (1 - \pi_I)(1 - F(0))} = \pi_I \geq \pi_C.$$

A high-type will never deviate from selecting the closest bounds - $\Psi(-\Psi)$ in equilibrium.

Consider now an uninformed type. Choosing some $x_1 \in (-\Psi, \Psi)$ is off-the-path, yielding a strictly worse payoff. By definition of $\bar{x}$, a low-type is indifferent between selecting $x_1 = \Psi$ and $x_1 = 0$. Moreover, the low-type is indifferent between choosing $-\Psi$ or Ψ as they are symmetric around 0. Therefore, the uninformed type mixes in equilibrium.

2. Suppose $\pi_I < \pi_C$. The updated belief that the incumbent is informed is

$$\tilde{\pi}_I(\Psi) = \frac{\pi_I(1 - F(0))}{\pi_I(1 - F(0)) + (1 - \pi_I)\tilde{\Sigma}},$$

where $\tilde{\Sigma}$ is the mixing probability of selecting $x_1 = \Psi$ for the low-type.

Given that $\pi_I < \pi_C$, the voter always prefers the challenger. As such, the low-type has an incentive to select $x_1 = 0$. For the uninformed type to mix, the voter must be indifferent. Let $\tilde{\pi}_I(\Psi) = \pi_C$ such that

$$\tilde{\pi}_I(\Psi) = \frac{\pi_I(1 - F(0))}{\pi_I(1 - F(0)) + (1 - \pi_I)\tilde{\Sigma}} = \pi_C.$$

Rearranging and,

$$\tilde{\Sigma} = \frac{\pi_I(1 - \pi_C)}{\pi_C(1 - \pi_I)}(1 - F(0)),$$

which hold as long as the low-type selects $x_1 = \Psi$ with probability $\frac{\pi_I(1 - \pi_C)}{\pi_C(1 - \pi_I)}(1 - F(0))$. An analogous case is made for $x_1 = -\Psi$.

All that is left is to determine the voter's mixing probability. Define the voter's continuation value for electing the challenger as

$$\begin{aligned} V_C(\Psi) = & \pi_C \left\{ \rho \left[\int_{-\infty}^{-\Psi} -(-\Psi - \omega_2)^2 f(\omega_2) d\omega_2 + \int_{\Psi}^{\infty} -(\Psi - \omega_2)^2 f(\omega_2) d\omega_2 \right] \right. \\ & \left. + (1 - \rho) \left[\int_{-\infty}^{-\Psi} -(-\Psi - \omega_2)^2 f(\omega_2) d\omega_2 + \int_{\Psi}^{\infty} -(\Psi - \omega_2)^2 f(\omega_2) d\omega_2 \right] \right\} \\ & + (1 - \pi_C) \left\{ \rho \left[\int_{-\infty}^{-\Psi} -(-\Psi - \omega_2)^2 f(\omega_2) d\omega_2 + \int_{\Psi}^{\infty} -(\Psi - \omega_2)^2 f(\omega_2) d\omega_2 \right] \right. \\ & \left. + (1 - \rho) \left[\int_{-\infty}^{\infty} -\omega_2^2 f(\omega_2) d\omega_2 \right] \right\}. \end{aligned}$$

For the low-type to mix, they need to be indifferent between selecting $x_1 = 0$ and

$x_1 = \Psi$. Therefore,

$$\begin{aligned}
\beta + V_C(\Psi) &= \beta - \sigma^2 - \Psi^2 + \mu(\Psi)[\beta - (1 - \rho)\sigma^2] + (1 - \mu(\Psi))V_C(\Psi) \\
\iff \beta &= \beta - \sigma^2 - \Psi^2 + \mu(\Psi)[\beta - (1 - \rho)\sigma^2] - \mu(\Psi)V_C(\Psi) \\
\iff \beta + \mu(\Psi)V_C(\Psi) &= \beta - \sigma^2 - \Psi^2 + \mu(\Psi)[\beta - (1 - \rho)\sigma^2] \\
\iff \mu(\Psi)V_C(\Psi) &= -\sigma^2 - \Psi^2 + \mu(\Psi)[\beta - (1 - \rho)\sigma^2] \\
\iff \mu(\Psi)V_C(\Psi) - \mu(\Psi)[\beta - (1 - \rho)\sigma^2] &= -\sigma^2 - \Psi^2 \\
\iff -\mu(\Psi)[\beta - (1 - \rho)\sigma^2 - V_C(\Psi)] &= -\sigma^2 - \Psi^2 \\
\iff \mu(\Psi) &= \frac{\Psi^2 + \sigma^2}{\beta - (1 - \rho)\sigma^2 - V_C(\Psi)}.
\end{aligned}$$

The uninformed type is indifferent between $-\Psi$ and Ψ because they are equidistant from 0 and $\mu(\Psi) = \mu(-\Psi)$. As before, all informed types, $\omega \in (-\Psi, \Psi)$ and $|\omega| > |\Psi|$, select the closest bound in equilibrium.

□

C.2.2 When $d_1 = P$.

Lemma C.2.2. (Symmetric Legislative Constraint when $d_1 = P$). *Suppose the first-period policy domain is particular ($d_1 = P$) and the legislative constraints are strong ($\Psi < \bar{x}$).*

1. *Assume the incumbent is popular ($\pi_I > \pi_C$).*

- *Informed Incumbent: If $\omega \geq 0$ then the incumbent chooses $x_1 = \Psi$. If $\omega < 0$, then the incumbent chooses $x_1 = -\Psi$.*
- *Uninformed Incumbent: The incumbent chooses $x_1 = \Psi$ with probability $1 - F(0)$ and $x = -\Psi$ with probability $F(0)$.*
- *Voter: The voter always re-elects the incumbent.*

2. *Assume the incumbent is unpopular ($\pi_I < \pi_C$).*

- *Informed Incumbent: If $\omega \geq 0$ then the incumbent chooses $x_1 = \Psi$. If $\omega < 0$, then the incumbent chooses $x_1 = -\Psi$.*
- *Uninformed Incumbent: The incumbent chooses $x_1 = \Psi$ with probability $\frac{\pi_I(1-\pi_C)}{\pi_C(1-\pi_I)}(1 - F(0))$ and $x = -\Psi$ with probability $\frac{\pi_I(1-\pi_C)}{\pi_C(1-\pi_I)}F(0)$.*
- *Voter: The voter reelects the incumbent with probability $\mu(\Psi) = \frac{\Psi^2}{\beta - (1-\rho)\sigma^2 - V_C(\Psi)}$.*

Proof of Lemma C.2.2.

1. Suppose $d_1 = P$ and $\pi_I > \pi_C$. The updated belief that the incumbent is high-type is

$$\tilde{\pi}_I(\Psi) = \frac{\pi_I(1 - F(0))}{\pi_I(1 - F(0)) + (1 - \pi_I)(1 - F(0))} = \pi_I \geq \pi_C.$$

A high-type will never deviate from selecting the closest bounds - $\Psi(-\Psi)$ in equilibrium.

Consider now an uninformed type. Choosing some $x_1 \in (-\Psi, \Psi)$ is off-the-path, yielding a strictly worse payoff. By definition of $\bar{x}$, a low-type is indifferent between selecting $x_1 = \Psi$ and $x_1 = 0$. Moreover, the low-type is indifferent between choosing $-\Psi$ or Ψ as they are symmetric around 0. Therefore, the uninformed type mixes in equilibrium.

2. Suppose $d_1 = P$ and $\pi_I < \pi_C$. The updated belief that the incumbent is informed is

$$\tilde{\pi}_I(\Psi) = \frac{\pi_I(1 - F(0))}{\pi_I(1 - F(0)) + (1 - \pi_I)\tilde{\Sigma}},$$

where $\tilde{\Sigma}$ is the mixing probability of selecting $x_1 = \Psi$ for the low-type.

Given that $\pi_I < \pi_C$, the voter always prefers the challenger. As such, the low-type has an incentive to select $x_1 = 0$. For the uninformed type to mix, the voter must be indifferent. Let $\tilde{\pi}_I(\Psi) = \pi_C$ such that

$$\tilde{\pi}_I(\Psi) = \frac{\pi_I(1 - F(0))}{\pi_I(1 - F(0)) + (1 - \pi_I)\tilde{\Sigma}} = \pi_C.$$

Rearranging and,

$$\tilde{\Sigma} = \frac{\pi_I(1 - \pi_C)}{\pi_C(1 - \pi_I)}(1 - F(0)),$$

which hold as long as the low-type selects $x_1 = \Psi$ with probability $\frac{\pi_I(1 - \pi_C)}{\pi_C(1 - \pi_I)}(1 - F(0))$. An analogous case is made for $x_1 = -\Psi$.

All that is left is to determine the voter's mixing probability. Define the voter's continuation value for electing the challenger as

$$\begin{aligned} V_C(\Psi) = & \pi_C \left\{ \rho \left[\int_{-\infty}^{-\Psi} -(-\Psi - \omega_2)^2 f(\omega_2) d\omega_2 + \int_{\Psi}^{\infty} -(\Psi - \omega_2)^2 f(\omega_2) d\omega_2 \right] \right. \\ & \left. + (1 - \rho) \left[\int_{-\infty}^{-\Psi} -(-\Psi - \omega_2)^2 f(\omega_2) d\omega_2 + \int_{\Psi}^{\infty} -(\Psi - \omega_2)^2 f(\omega_2) d\omega_2 \right] \right\} \\ & + (1 - \pi_C) \left\{ \rho \left[\int_{-\infty}^{-\Psi} -(-\Psi - \omega_2)^2 f(\omega_2) d\omega_2 + \int_{\Psi}^{\infty} -(\Psi - \omega_2)^2 f(\omega_2) d\omega_2 \right] \right. \\ & \left. + (1 - \rho) \left[\int_{-\infty}^{\infty} -\omega_2^2 f(\omega_2) d\omega_2 \right] \right\}. \end{aligned}$$

For the low-type to mix, they need to be indifferent between selecting $x_1 = 0$ and

$x_1 = \Psi$. Therefore,

$$\begin{aligned}
\beta - \sigma^2 + V_C(\Psi) &= \beta - \sigma^2 - \Psi^2 + \mu(\Psi)[\beta - (1 - \rho)\sigma^2] + (1 - \mu(\Psi)) V_C(\Psi) \\
\iff \beta - \sigma^2 &= \beta - \sigma^2 - \Psi^2 + \mu(\Psi)[\beta - (1 - \rho)\sigma^2] - \mu(\Psi) V_C(\Psi) \\
\iff \beta - \sigma^2 + \mu(\Psi) V_C(\Psi) &= \beta - \sigma^2 - \Psi^2 + \mu(\Psi)[\beta - (1 - \rho)\sigma^2] \\
\iff \mu(\Psi) V_C(\Psi) &= -\Psi^2 + \mu(\Psi)[\beta - (1 - \rho)\sigma^2] \\
\iff \mu(\Psi) V_C(\Psi) - \mu(\Psi)[\beta - (1 - \rho)\sigma^2] &= -\Psi^2 \\
\iff -\mu(\Psi)[\beta - (1 - \rho)\sigma^2 - V_C(\Psi)] &= -\Psi^2 \\
\iff \mu(\Psi) &= \frac{\Psi^2}{\beta - (1 - \rho)\sigma^2 - V_C(\Psi)}.
\end{aligned}$$

The uninformed type is indifferent between $-\Psi$ and Ψ because they are equidistant from 0 and $\mu(\Psi) = \mu(-\Psi)$. As before, all informed types - $\omega \in (-\Psi, \Psi)$ and $|\omega| > |\Psi|$ - select the closest bound in equilibrium.

□

Proposition 7 (Impact of ρ on Voters' Perceptions). *As ρ increases, the re-election probability for the uninformed incumbent increases.*

Proof of Proposition 7. Define the total derivative as

$$\frac{d\mu(\Psi^U)}{d\rho} = \frac{\partial\mu(\Psi^U)}{\partial\rho} + \left[\frac{\partial\mu(\Psi^U)}{\partial\mathbb{E}_{U_I}(x(\Psi^U))} \cdot \frac{\partial\mathbb{E}_{U_I}(x(\Psi^U))}{\partial\rho} + \frac{\partial\mu(\Psi^U)}{\partial V_C(\Psi^U)} \cdot \frac{\partial V_C(\Psi^U)}{\partial\rho} \right].$$

There is no direct effect, so

$$\frac{\partial\mu(\Psi^U)}{\partial\rho} = 0.$$

To locate the indirect effect:

1. **Indirect effect of $\mathbb{E}_{U_I}(x(\Psi^U))$.** Let

$$D := \beta - (1 - \rho)\sigma^2 - V_C(\Psi^U), \quad N := \Psi^2 + \sigma^2.$$

Then

$$\frac{\partial\mu(\Psi^U)}{\partial\mathbb{E}_{U_I}(x(\Psi^U))} = \frac{D \cdot N' - N \cdot D'}{D^2} = \frac{\Psi^2 + \sigma^2}{[\beta - (1 - \rho)\sigma^2 - V_C(\Psi^U)]^2} > 0.$$

Also,

$$\frac{\partial\mathbb{E}_{U_I}(x(\Psi^U))}{\partial\rho} = \frac{\partial}{\partial\rho} [-(1 - \rho)\sigma^2] = \sigma^2 > 0.$$

2. **Indirect effect of $V_C(\Psi^U)$.** With the same D, N as above,

$$\frac{\partial \mu(\Psi^U)}{\partial V_C(\Psi^U)} = \frac{\Psi^2 + \sigma^2}{[\beta - (1 - \rho)\sigma^2 - V_C(\Psi^U)]^2}.$$

Define

$$A(\Psi^U) := \int_{-\infty}^{-\Psi^U} -(\Psi^U - \omega)^2 f(\omega) d\omega + \int_{\Psi^U}^{\infty} -(\Psi^U - \omega)^2 f(\omega) d\omega,$$

and

$$B := \int_{-\infty}^{\infty} -\omega^2 f(\omega) d\omega = -\sigma^2.$$

Then

$$V_C(\Psi^U) = \pi_C [\rho A(\Psi^U) + (1 - \rho)A(\Psi^U)] + (1 - \pi_C) [\rho A(\Psi^U) + (1 - \rho)B].$$

Thus

$$\frac{dV_C(\Psi^U)}{d\rho} = \pi_C [A(\Psi^U) - A(\Psi^U)] + (1 - \pi_C) [A(\Psi^U) - B] = (1 - \pi_C) [A(\Psi^U) - B].$$

By change of variables,

$$A(\Psi^U) = 2 \int_{\Psi^U}^{\infty} -(\omega - \Psi^U)^2 f(\omega) d\omega,$$

so

$$\frac{dV_C(\Psi^U)}{d\rho} = (1 - \pi_C) \left[2 \int_{\Psi^U}^{\infty} -(\omega - \Psi^U)^2 f(\omega) d\omega - (-\sigma^2) \right] > 0.$$

3. **Total derivative.** Substituting the components,

$$\begin{aligned} \frac{d\mu(\Psi^U)}{d\rho} &= \frac{\Psi^2 + \sigma^2}{[\beta - (1 - \rho)\sigma^2 - V_C(\Psi^U)]^2} \left\{ \sigma^2 + (1 - \pi_C) \left[2 \int_{\Psi^U}^{\infty} -(\omega - \Psi^U)^2 f(\omega) d\omega + \sigma^2 \right] \right\} \\ &> 0. \end{aligned}$$

□

D Ideology

I extend the game to a setting in which politicians have an ideological bias, L or R. While the previous analyses provide clear insights into how contextual knowledge structures incumbents' incentives, polarization is an important factor to consider when thinking about

voter perceptions and racial competency. Incumbents who lack racial competency may polarize their policy position, assuming that particularized needs can be met with ideological extremity. As in Bils (2023), to introduce polarization, assume that the incumbent has some bias R and the challenger a bias L , with $L < 0 < R$. The updated payoff for the incumbent is $-(x - \omega_t - R)^2$, while the voter's payoff remains $-(x - \omega_t)^2$. The reduced-form expressions for the upper and lower bounds of bias R , and the state threshold are

$$\begin{aligned}\bar{R} &= \sqrt{L^2 + (1 - \pi)(1 - \rho)\sigma^2}, \\ \underline{R} &= \sqrt{\max\{0, L^2 - (\rho + \pi - \pi\rho)\sigma^2\}}, \\ \omega_R^P &= \sqrt{\beta + \rho\sigma^2 - \pi\sigma^2 + (R - L)^2}.\end{aligned}$$

Furthermore, I normalize the prior beliefs about the incumbent and challenger's competency such that $\pi_I = \pi_C = \pi$. Polarization is defined as the difference between L and R .

I begin by briefly analyzing lopsided elections. To define lopsided elections, assume that the voter has some preference for the incumbent who is closer to their ideological position. Then, for some $R < \underline{R}$, the voter will re-elect the incumbent even if they are uninformed; conversely, if $R > \bar{R}$, the challenger always wins even if the incumbent is racially competent.

In lopsided elections, both sets of politicians select their first-best option, given that the election is a foregone conclusion. What matters here is that ρ factors into the upper and lower bounds of the incumbent's ideological extremity. As ρ increases, the upper bound decreases while the lower bound approaches 0. Otherwise put, the incumbent has to polarize less to maintain their lopsided victory when racial competence is not required. This has important implications for competitive elections, where both the willingness and threshold to distort are central.

Unlike a lopsided election, competitive settings introduce the incentive to distort policy again. Let $\underline{R} < R < \bar{R}$ define a competitive election. Furthermore, let $\bar{x}_R^P = R + \omega_R^P$ and $\underline{x}_R^P = R - \omega_R^P$. I define the mixing probability for the uninformed type as:

$$\begin{aligned}\bar{\Sigma}_R &= \frac{\pi}{1 - \pi} \frac{1 - \pi - \frac{R^2 - L^2}{\sigma^2}}{\pi + \frac{R^2 - L^2}{\sigma^2}} (F(\omega_R^P) - F(0)), \\ \underline{\Sigma}_R &= \frac{\pi}{1 - \pi} \frac{1 - \pi - \frac{R^2 - L^2}{\sigma^2}}{\pi + \frac{R^2 - L^2}{\sigma^2}} (F(0) - F(-\omega_R^P)).\end{aligned}$$

Here, racial competency operates directly through the condition for what makes an election competitive, but also indirectly through the state threshold. Moreover, racial competency directly affects the incumbents' threshold for policy extremity. Therefore, the need for

contextual knowledge affects not only the incumbent’s winning probability but also how both polarization and contextual knowledge influence the extremity of policy distortion.

Lemma A1 defines the equilibrium for competitive elections.

Lemma A1 (Competitive Elections). *Suppose the election is competitive. Then,*

1. *There is a perfect Bayesian equilibrium analogous to Proposition 1.*
2. *As polarization increases, so does ω_R^P .*

When elections are competitive, voters are willing to re-elect the incumbent as long as $|x_1| \geq \bar{x}_R^P$. This creates an incentive for the incumbent to overreact or posture strategically to secure re-election. Importantly, the threshold for policy distortion, ω_R^P , rises with polarization. Here, racial competency operates by affecting both polarization and the threshold for policy distortion. As racial competence becomes less important tomorrow, the threshold for which the incumbent can exaggerate policy widens; yet, ρ also narrows the bounds at which the election is competitive, circumscribing the incumbent to a set of smaller R required to win.

As Bils (2023) notes, asymmetric polarization makes it difficult to characterize the incumbent’s mixing behavior. Competitive elections can become lopsided when one politician diverges at a faster rate, making it unclear how racial competency is structuring behavior. To facilitate comparative statics, I therefore assume symmetric polarization by letting the incumbent and challenger’s bias diverge at equal rates. This assumption clarifies the role of racial competency by isolating its effect on the incumbent’s winning probability and the extremity of distortion.

Proposition A1 (Comparative Statics Under Symmetric Polarization). *Suppose the incumbent’s and challenger’s biases are equidistant ($R = -L$). Then,*

1. **Effect of polarization.** *The incumbent’s winning probability increases in polarization:*

$$\frac{\partial P_{win}^L}{\partial R} > 0.$$

2. **Effect of racial competency.** *The incumbent’s winning probability increases in the probability of a universal second period:*

$$\frac{\partial P_{win}^L}{\partial \rho} > 0.$$

3. **Interaction effect on the state threshold.** *The marginal benefit of polarization on the state threshold decreases as ρ increases:*

$$\frac{\partial^2 \omega_R^P}{\partial R \partial \rho} < 0.$$

4. **Interaction effect on the winning probability.** *The marginal benefit of polarization on the incumbent’s winning probability decreases as ρ increases:*

$$\frac{\partial^2 P_{win}^L}{\partial R \partial \rho} < 0.$$

Proposition A1 establishes that under symmetric polarization, the winning probability for the uninformed type increases as polarization increases. Because polarizing the election increases the threshold for policy distortion, both types of incumbents are more willing to overreact or posture in equilibrium. When we examine the effect of the need for contextual knowledge in the next period, a similar pattern emerges. Racial competency affects the incumbent’s mixing probability by increasing their willingness to posture in equilibrium. As the probability that the next period becomes universal increases, the incumbent faces a greater incentive to mix. Therefore, their winning probability also rises, as depicted in part two of Proposition A1.

Polarization and racial competency are substitutes in the electoral arena. If the Multiracial incumbent is unable to deliver effective, particularized policy in the next period, they can polarize and secure re-election. Similarly, a reduction in the salience of contextual knowledge has a similar effect by loosening the voter’s standard for racial competence. This equivalence is summarized in the following corollary.

Corollary A1 (Electoral Equivalence). *ρ and R are electorally equivalent under symmetric polarization.*

While they are electoral substitutes, the incumbent cannot know ex ante which behavior to adopt to secure re-election. Uncertainty about contextual knowledge may drive the incumbent to polarize now to improve perceptions of their racial competence; however, this may also stifle them with voters, as voters are less willing to tolerate polarizing elections when the next period is universal. The bounds of competitive elections shrink as ρ increases.

Part 3 of Proposition A1 begins to unpack the joint effect of polarizing and the need for racial competency on the incumbent’s policy distortion threshold. As the probability that ρ increases, polarizing the election leads to a constraint on the extremity of their policy. At first glance, this appears to contradict the earlier claim that universal periods lead to more posturing. The distinction, however, is that this highlights a marginal effect: for a one-unit increase in polarization, the effect on shifting the policy distortion threshold diminishes. Moreover, this connects to a theme seen throughout the ideological extension—voters require less polarization in universal periods. This phenomenon was found in lopsided and asymmetric competitive elections. Incumbents are better off following their first-best behavior rather than polarizing. But, because they want to be perceived as racially competent, they polarize to safeguard against the possibility that the second period requires contextual knowledge. This effect has important empirical implications. In Davenport (2016) analysis of Multiracial attitudes, she finds that on issues related to social policy, mixed-race respondents are more liberal than their Monoracial minority peers. Implication A1 provides a mechanism for this pattern.

Implication A1. *For Multiracial incumbents, when racial competency is required, uninformed types are more likely to polarize.*

Because racial competence affects not only the bounds of policy distortion but the mixing probability, the need for contextual knowledge both directly and indirectly affects the incumbent’s winning probability. Part 4 of Proposition A1 shows that as polarization and the probability of a universal period rise together, the incumbent’s advantage shrinks. Again, the same story is clear: the incumbent engages in less policy distortion, which makes it easier for voters to screen out racially incompetent types in equilibrium. Polarizing is beneficial to the incumbent only when the next period demands racial competence, since it blurs voters’ perceptions.

Earlier, I provided evidence that perceptions of racial competency heighten the pressure to act within group constraints. Substantively, this was due to the role of reflected appraisals, the process by which Multiracial identity is defined through others’ perceptions, in guiding Multiracial behavior (Khanna 2004, 2010; Sims 2016). Here, polarization provides an avenue for Multiracial incumbents to blur the lines of racial competence. Therefore, polarization serves not only as an important conduit of perceptions but also diminishes the burden of evidence for Multiracial incumbents. Implication A2 connects reflected appraisals to the work by Davenport (2016) and Davenport, Franco, and Iyengar (2022), showing why uninformed Multiracials are more likely to polarize to appear racially competent.

Implication A2. *For Multiracial incumbents, polarization acts as a mechanism of reflected appraisal: by exaggerating policy extremity, uninformed types reduce the burden of proving racial competency.*

D.1 Proofs to Extension: Ideology

Lemma A1. (Competitive Elections). *Suppose the election is competitive. Then,*

1. *There is a perfect Bayesian equilibrium analogous to Proposition 1.*
2. *As polarization increases, so does ω_R^P .*

Proof of Lemma A1. The first part of Lemma A1 follows from Proposition 1.

1. Uninformed Incumbent.

Suppose $\pi_I = \pi_C = \pi$. The uninformed type’s utility is

$$\beta - \sigma^2 - (R - L)^2 - (1 - \pi)\sigma^2,$$

For selecting R. For $\bar{x}_R^P$ (analogous case for $\underline{x}_R^P$),

$$\beta - \sigma^2 - (\bar{x}_R^P - L)^2 - (1 - \pi)\sigma^2.$$

Choosing any $x_1 \in (\underline{x}_R^P, \bar{x}_R^P)$ yields a payoff that is weakly worse than choosing $x_1 = R$. Moreover, selecting $x_1 > \bar{x}_R^P$ or $x_1 < \underline{x}_R^P$ is strictly worse than choosing $x_1 = \bar{x}_R^P$ or $x_1 = \underline{x}_R^P$. Therefore, the uninformed incumbent is indifferent between $x_1 > \bar{x}_R^P$, $x_1 < \underline{x}_R^P$, and $x_1 = R$. The uninformed type mixes in equilibrium.

2. Informed Incumbent.

For the informed type, if $\omega_1 > \bar{x}_R^P$ or $\omega_1 < \underline{x}_R^P$, the optimal policy is to choose $x_1 = \omega_1 + R$. Now partition the state space such that for $\omega_1 \in [0, \bar{x}_R^P]$, the optimal policy for re-election is $x_1 = \bar{x}_R^P$. If the informed type selected $x_1 = \omega_1 + R$, they would be removed from office. Thus, their payoffs are

$$\beta - (1 - \pi)\sigma^2 - (L - R)^2,$$

if removed. And,

$$-(\bar{x}_R^P - \omega_1)^2 + 2\beta,$$

if re-elected. The incumbent will not deviate from $\bar{x}_R^P$ if

$$\begin{aligned} -(x_R^P)^2 + 2\beta &> \beta - (1 - \pi)\sigma^2 - (L - R)^2 \\ -(x_R^P)^2 + \beta &> -(1 - \pi)\sigma^2 - (L - R)^2 \\ \beta + (1 - \pi)\sigma^2 + (L - R)^2 &> (x_R^P)^2 \\ \sigma^2 - \pi\sigma^2 &> \rho\sigma^2 - \pi\sigma^2 \\ \sigma^2 &> \rho\sigma^2. \end{aligned}$$

where $(x_R^P)^2 = \beta + \rho\sigma^2 - \pi\sigma^2 + (R - L)^2$. This inequality holds for $\rho \in [0, 1)$. A similar argument exists for $\omega_1 \in (\underline{x}_R^P, 0)$. Thus, the $\omega_1 = 0$ type prefers $x_1 = \bar{x}_R^P$ over $x_1 = \omega_1 + R$.

3. Voter Decision.

When $x_1 = \bar{x}_R^P$ it is required that $\tilde{\pi}_I(x_1) \geq \pi_C$ for the incumbent to be re-elected. The voter's expected utility for re-electing the incumbent is

$$-R^2 - (1 - \tilde{\pi}_I(x_1))\sigma^2,$$

and for electing the challenger,

$$-L^2 - (1 - \pi_C)\sigma^2.$$

Thus,

$$\begin{aligned}
-R^2 - (1 - \tilde{\pi}_I(x_1))\sigma^2 &\geq -L^2 - (1 - \pi_C)\sigma^2 \\
-R^2 - \sigma^2 + \tilde{\pi}_I(x_1)\sigma^2 &\geq -L^2 - \sigma^2 + \pi_C\sigma^2 \\
\tilde{\pi}_I(x_1) &\geq \pi_C + \frac{R^2 - L^2}{\sigma^2},
\end{aligned}$$

where my definition of a competitive election $0 < R^2 - L^2 < 1$. By a slightly revised version of Lemma 2,

$$\tilde{\pi}_I(x_1) = \frac{(F(\omega_R^P) - F(0)) \cdot \pi_I}{(F(\omega_R^P) - F(0)) \cdot \pi_I - (1 - \pi_C)\bar{\Sigma}_R} \geq \pi_C + \frac{R^2 - L^2}{\sigma^2}.$$

Let $A := F(\omega_R^P)$, $B := F(0)$, $\Delta = \frac{R^2 - L^2}{\sigma^2}$. Then,

$$\begin{aligned}
\tilde{\pi}_I(x_1) &= \frac{(A - B) \cdot \pi}{(A - B) \cdot \pi - (1 - \pi)\bar{\Sigma}_R} \geq \pi + \Delta \\
(A - B)\pi &\geq (\pi + \Delta)[\pi(A - B) + (1 - \pi)\bar{\Sigma}_R] \\
(A - B)\pi &\geq (\pi + \Delta)(\pi(A - B)) + (\pi + \Delta)(1 - \pi)\bar{\Sigma}_R \\
(A - B)\pi - \pi(A - B)(\pi + \Delta) &\geq (\pi + \Delta)(1 - \pi)\bar{\Sigma}_R \\
(A - B)\pi[1 - (\pi + \Delta)] &\geq (\pi + \Delta)(1 - \pi)\bar{\Sigma}_R \\
\frac{(A - B)\pi[1 - \pi - \Delta]}{(\pi + \Delta)(1 - \pi)} &\geq \bar{\Sigma}_R \\
\frac{\pi}{1 - \pi} \cdot \frac{1 - \pi - \Delta}{\pi + \Delta} &\geq \bar{\Sigma}_R.
\end{aligned}$$

A similar argument holds for $x_1 = \underline{x}_R^P$.

The second part of Lemma A1 comes from differentiating ω_R^P WRT $R - L$.

□

Proposition A1. (Comparative Statics Under Symmetric Polarization). *Suppose the incumbent's and challenger's biases are equidistant ($R = -L$). Then,*

1. **Effect of polarization.** The incumbent's winning probability increases in polarization:

$$\frac{\partial P_{win}^L}{\partial R} > 0.$$

2. **Effect of racial competency.** The incumbent's winning probability increases in the probability of a universal second period:

$$\frac{\partial P_{win}^L}{\partial \rho} > 0.$$

3. **Interaction effect on the state threshold.** The marginal benefit of polarization on the state threshold decreases as ρ increases:

$$\frac{\partial^2 \omega_R^P}{\partial R \partial \rho} < 0.$$

4. **Interaction effect on the winning probability.** The marginal benefit of polarization on the incumbent's winning probability decreases as ρ increases:

$$\frac{\partial^2 P_{win}^L}{\partial R \partial \rho} < 0.$$

Proof of Proposition A1.

1. To show $\frac{\partial P_{win}^L}{\partial R} > 0$.

Suppose that the politicians are symmetrically polarized such that $R = -L$. Further, define the winning probability for the low-type as

$$P_{win}^L := \pi + (1 - \pi)[\bar{\Sigma}_R + \underline{\Sigma}_R].$$

We want to show that $\frac{\partial P_{win}^L(R)}{\partial R} > 0$. First, expand the mixing probability such that

$$\pi + (1 - \pi) \left[\frac{\pi}{1 - \pi} \frac{1 - \pi - \frac{(R^2 - (R^2))}{\sigma^2}}{\pi + \frac{(R^2 - (R^2))}{\sigma^2}} \right] (F(\omega_R^P) - F(-\omega_R^P)).$$

The mixing probability simplifies to

$$\pi + (1 - \pi)(F(\omega_R^P) - F(-\omega_R^P)).$$

Next, to solve the partial derivative with respect to R , we utilize the chain rule.

$$\frac{\partial P_{win}^L}{\partial R} = (1 - \pi) \left[f(\omega_R^P) \frac{\partial \omega_R^P}{\partial R} - f(-\omega_R^P) \frac{\partial (-\omega_R^P)}{\partial R} \right].$$

Because $\frac{\partial (-\omega_R^P)}{\partial R} = -1 \left(\frac{\partial (\omega_R^P)}{\partial R} \right)$,

$$\frac{\partial P_{win}^L}{\partial R} = (1 - \pi) \left[f(\omega_R^P) \frac{\partial \omega_R^P}{\partial R} + f(-\omega_R^P) \frac{\partial (\omega_R^P)}{\partial R} \right].$$

The sign of the winning probability is dependent on $\frac{\partial(\omega_R^P)}{\partial R}$. Let $\omega_R^P = \sqrt{\beta + \rho\sigma^2 - \pi\sigma^2 + (R - L)^2}$. Because of symmetric polarization, $\omega_R^P = \sqrt{\beta + \rho\sigma^2 - \pi\sigma^2 + 4R^2}$. To solve WRT R , we utilize the chain rule:

$$\frac{\partial\omega_R^P}{\partial R} = f'(g(R)) \cdot g'(R),$$

where $g(R) = \beta + \rho\sigma^2 - \pi\sigma^2 + 4R^2$. First,

$$f'(g(R)) = \frac{1}{2}(g(R))^{-\frac{1}{2}}.$$

Second,

$$g'(R) = 8R.$$

Combining and simplifying results in

$$\frac{\partial\omega_R^P}{\partial R} = \frac{1}{2}(g(R))^{-\frac{1}{2}} \cdot 8R = \frac{4R}{\omega_R^P}.$$

Because $\frac{\partial\omega_R^P}{\partial R} > 0$, $\frac{\partial P_{win}^L}{\partial R} > 0$. Therefore, as polarization increases, so does the winning probability for the low-type.

2. To show $\frac{\partial P_{win}^L}{\partial \rho} > 0$.

Suppose that the politicians are symmetrically polarized such that $R = -L$. Further, define the winning probability for the low-type as

$$P_{win}^L := \pi + (1 - \pi)[\bar{\Sigma}_R + \underline{\Sigma}_R].$$

We want to show that $\frac{\partial P_{win}^L(R)}{\partial \rho} > 0$. Expand the mixing probability such that

$$\pi + (1 - \pi) \left[\frac{\pi}{1 - \pi} \frac{1 - \pi - \frac{(R^2 - (R^2))}{\sigma^2}}{\pi + \frac{(R^2 - (R^2))}{\sigma^2}} \right] (F(\omega_R^P) - F(-\omega_R^P)).$$

The mixing probability simplifies to

$$\pi + (1 - \pi)(F(\omega_R^P) - F(-\omega_R^P)).$$

Next, to solve the partial derivative with respect to ρ , we utilize the chain rule.

$$\frac{\partial P_{win}^L}{\partial \rho} = (1 - \pi) \left[f(\omega_R^P) \frac{\partial \omega_R^P}{\partial \rho} - f(-\omega_R^P) \frac{\partial (-\omega_R^P)}{\partial \rho} \right].$$

Because $\frac{\partial(-\omega_R^P)}{\partial \rho} = -1 \left(\frac{\partial(\omega_R^P)}{\partial \rho} \right)$,

$$\frac{\partial P_{win}^L}{\partial \rho} = (1 - \pi) \left[f(\omega_R^P) \frac{\partial \omega_R^P}{\partial \rho} + f(-\omega_R^P) \frac{\partial (\omega_R^P)}{\partial \rho} \right].$$

The sign of the winning probability is dependent on $\frac{\partial(\omega_R^P)}{\partial\rho}$. Let $\omega_R^P = \sqrt{\beta + \rho\sigma^2 - \pi\sigma^2 + (R - L)^2}$. Because of symmetric polarization, $\omega_R^P = \sqrt{\beta + \rho\sigma^2 - \pi\sigma^2 + 4R^2}$. To solve WRT ρ , we utilize the chain rule:

$$\frac{\partial\omega_R^P}{\partial\rho} = f'(g(\rho)) \cdot g'(\rho),$$

where $g(\rho) = \beta + \rho\sigma^2 - \pi\sigma^2 + 4R^2$. First,

$$f'(g(\rho)) = \frac{1}{2}(g(\rho))^{-\frac{1}{2}}.$$

Second,

$$g'(\rho) = \sigma^2.$$

Combining and simplifying results in

$$\frac{\partial\omega_R^P}{\partial\rho} = \frac{1}{2}g(\rho)^{-\frac{1}{2}} \cdot \sigma^2 = \frac{\sigma^2}{2\omega_R^P}.$$

Because $\frac{\partial\omega_R^P}{\partial\rho} > 0$, $\frac{\partial P_{win}^L}{\partial\rho} > 0$. Therefore, as the probability that the next period is universal increases, so does the winning probability for the low-type.

3. To show $\frac{\partial^2\omega_R^P}{\partial R\partial\rho} < 0$.

Given that

$$\frac{\partial\omega_R^P}{\partial\rho} = \frac{\sigma^2}{2\omega_R^P},$$

the cross-partial is

$$\frac{\partial^2\omega_R^P}{\partial R\partial\rho} = \frac{\partial[\frac{\sigma^2}{2\omega_R^P}]}{\partial R}.$$

I want to show that

$$\frac{\partial^2\omega_R^P}{\partial R\partial\rho} < 0.$$

By the quotient rule, let $N = \sigma^2$ and $D = 2\omega_R^P$. Then,

$$\begin{aligned} \frac{\partial^2\omega_R^P}{\partial R\partial\rho} &= \frac{D \cdot N' - N \cdot D'}{D^2} \\ &= \frac{2\omega_R^P \cdot 0 - \sigma^2 \cdot \frac{\partial D}{\partial R}}{(2\omega_R^P)^2} \\ &= -\frac{2\sigma^2}{4(\omega_R^P)^2} \cdot \frac{\partial\omega_R^P}{\partial R} \\ &= -\frac{\sigma^2}{2(\omega_R^P)^2} \cdot \frac{4R}{\omega_R^P} \\ &= -\frac{2\sigma^2 R}{(\omega_R^P)^3}. \end{aligned}$$

Therefore,

$$\frac{\partial^2 \omega_R^P}{\partial R \partial \rho} < 0,$$

implying that the marginal benefit of widening polarization on the state threshold decreases as ρ increases.

4. To show $\frac{\partial^2 P_{win}^L}{\partial R \partial \rho} < 0$.

Suppose that the politicians are symmetrically polarized such that $R = -L$. Further, define the winning probability for the low-type as

$$P_{win}^L := \pi + (1 - \pi)[\bar{\Sigma}_R + \underline{\Sigma}_R].$$

We want to show that $\frac{\partial^2 P_{win}^L(R)}{\partial R \partial \rho} < 0$. Expand the mixing probability such that

$$\pi + (1 - \pi) \left[\frac{\pi}{1 - \pi} \frac{1 - \pi - \frac{(R^2 - (R^2))}{\sigma^2}}{\pi + \frac{(R^2 - (R^2))}{\sigma^2}} \right] (F(\omega_R^P) - F(-\omega_R^P)).$$

The mixing probability simplifies to

$$\pi + (1 - \pi)(F(\omega_R^P) - F(-\omega_R^P)).$$

Next, we know that

$$\frac{\partial P_{win}^L(R)}{\partial \rho} = (1 - \pi)[f(\omega_R^P) + f(-\omega_R^P)] \frac{\partial \omega_R^P}{\partial \rho}.$$

Now, to locate $\frac{\partial^2 P_{win}^L(R)}{\partial R \partial \rho}$,

$$\frac{\partial^2 P_{win}^L(R)}{\partial R \partial \rho} = \frac{\partial [(1 - \pi)[f(\omega_R^P) + f(-\omega_R^P)] \frac{\partial \omega_R^P}{\partial \rho}]}{\partial R}.$$

Let $A := (1 - \pi)[f(\omega_R^P) + f(-\omega_R^P)]$ and $B := \frac{\partial \omega_R^P}{\partial \rho}$. First,

$$\begin{aligned} \frac{\partial A}{\partial R} &= (1 - \pi)[f'(\omega_R^P) \frac{\partial \omega_R^P}{\partial R} + f'(-\omega_R^P) \frac{\partial (-\omega_R^P)}{\partial R}] \\ &= (1 - \pi)[f'(\omega_R^P) - f'(-\omega_R^P)] \frac{\partial \omega_R^P}{\partial R}. \end{aligned}$$

Second,

$$\frac{\partial B}{\partial R} = \frac{\partial [\frac{\partial \omega_R^P}{\partial \rho}]}{\partial R} = \frac{\partial^2 \omega_R^P}{\partial R \partial \rho},$$

which is equal to $-\frac{2\sigma^2 R}{(\omega_R^P)^3}$. By the product rule,

$$\frac{\partial^2 P_{win}^L(R)}{\partial R \partial \rho} = \frac{\partial A}{\partial R} \cdot B + A \frac{\partial B}{\partial R}$$

$$= [(1 - \pi)[f'(\omega_R^P) - f'(-\omega_R^P)] \frac{\partial \omega_R^P}{\partial R}] \frac{\partial \omega_R^P}{\partial \rho} + [(1 - \pi)[f(\omega_R^P) + f(-\omega_R^P)]] \frac{\partial^2 \omega_R^P}{\partial R \partial \rho}.$$

Rearrange and simplify,

$$(1 - \pi) \left\{ [f(\omega_R^P) + f(-\omega_R^P)] \frac{\partial^2 \omega_R^P}{\partial R \partial \rho} + [f'(\omega_R^P) - f'(-\omega_R^P)] \frac{\partial \omega_R^P}{\partial R} \frac{\partial \omega_R^P}{\partial \rho} \right\}.$$

We know that

$$\frac{\partial \omega_R^P}{\partial R} > 0 \quad \frac{\partial \omega_R^P}{\partial \rho} > 0,$$

and,

$$\frac{\partial^2 \omega_R^P}{\partial R \partial \rho} < 0.$$

Therefore,

$$\frac{\partial^2 P_{win}^L(R)}{\partial R \partial \rho} < 0,$$

implying that the marginal benefit of widening polarization on the winning probability decreases as ρ increases.

□

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